My chart is Lying !

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WOMEN IN DATA SCIENCE
STANFORD UNIVERSITY

Meet the presenter

Dr Nisha is an educator by heart & a trainer by profession

Learning new skill sets excites her & she is fanatical about knowledge sharing.

Her online profiles speak about her contributions to the learning/ research Communit.

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Background

- PhD in Mathematics.
- Works in the area of Data Science, Statistical Research, Data Visualization & Storytelling
- Creator of various courses
- Contributor to various research communities and Q/A forums
- Mentor/Panelist for women in Tech Global

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Her Contribution to the Community

http://stats.stackexchange.com/users/79100/learner

https://stackoverflow.com/users/5114585/dr-nisha-arora

https://www.quora.com/profile/Nisha-Arora-9

https://www.researchgate.net/profile/Nisha_Arora2/contributions

http://learnerworld.tumblr.com/

https://www.slideshare.net/NishaArora1

https://scholar.google.com/citations?user=JgCRWh4AAAAJ&hl=en&authuser=1

https://www.youtube.com/channel/UCniyhvrD_8AM2jXki3eEErw

https://groups.google.com/g/dataanalysistraining/search?q=nisha%20arora

https://www.linkedin.com/in/drnishaarora/detail/recent-activity/posts/

- ✓ Research Queries
- ✓ Coding Queries
- ✓ Blog Posts
- ✓ Slide Decks
- ✓ My Talks
- ✓ Publications
- ✓ Lectures
- ✓ Layman's Term Explanation
- ✓ Mentoring
- ✓ Articles & Much More

Breakout questions

1

How to read a chart?

2

Do you think there is a lot for misinformation & lies spread by charts everywhere?

3

How do you handle it?

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Agenda

01 Common Myths

02 Catch the lies, if you can!

03 Know the Rules!

04 What's next _ Technically correct yet misleading charts

05 Closing

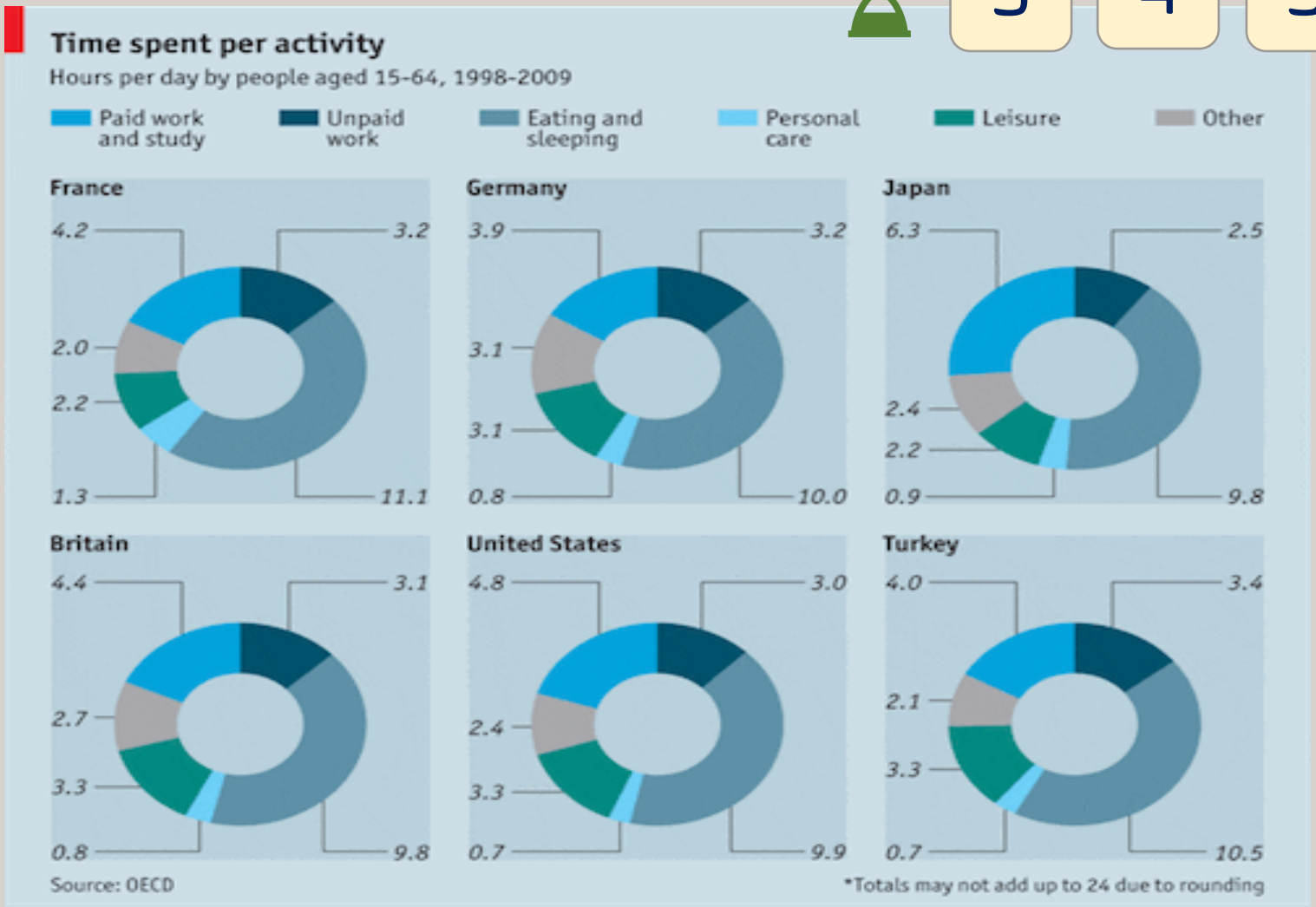
01

Common Myths

A Picture is worth a thousand words.

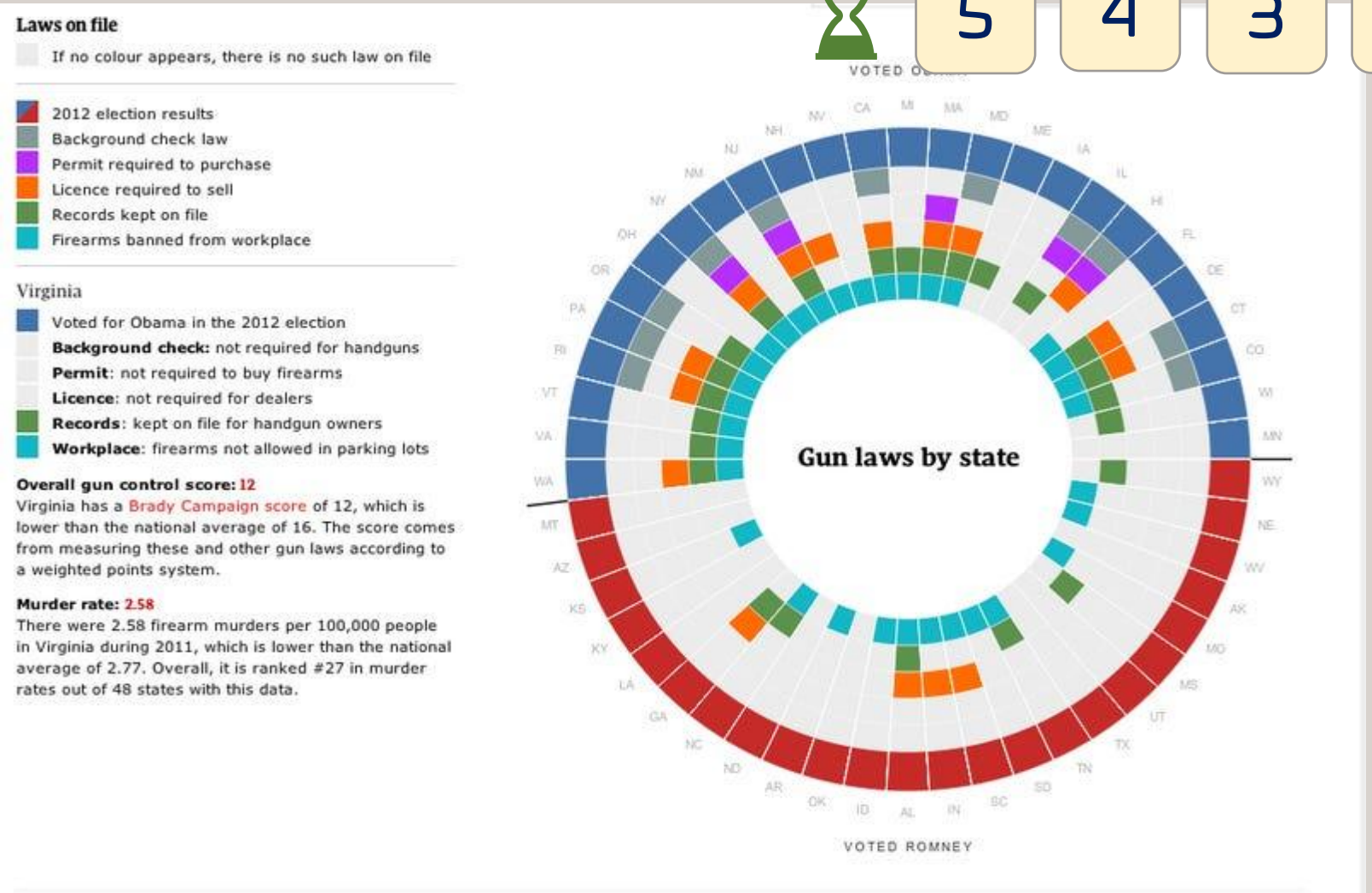
Not Really!

5 4 3 2 1 0



Too Much Cognitive Load!

5 4 3 2 1 0

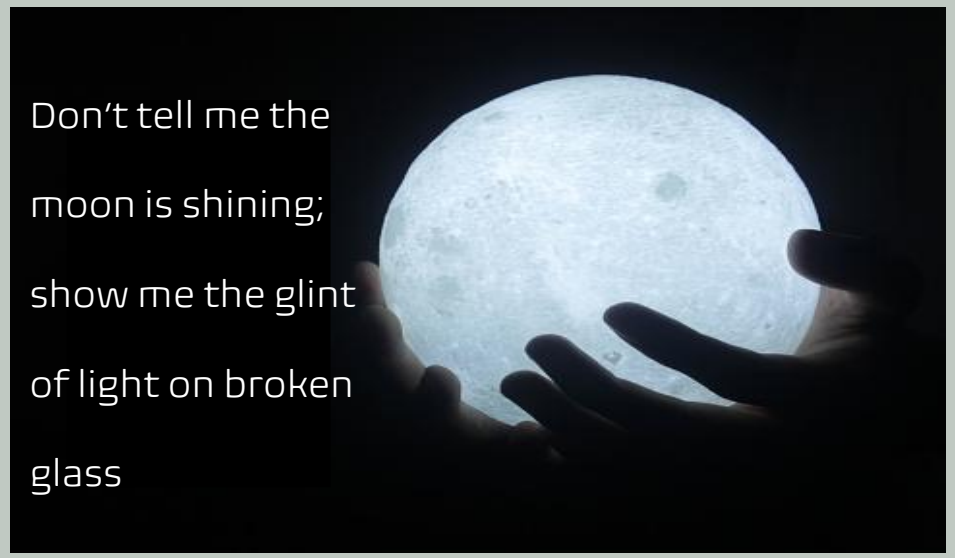


Don’t know how to read it!

A Picture is worth a thousand words, only if

- ✓ You know how to read it
- ✓ It is conveying the right information effortlessly to a wider audience
- ✓ It’s not biased or misleading

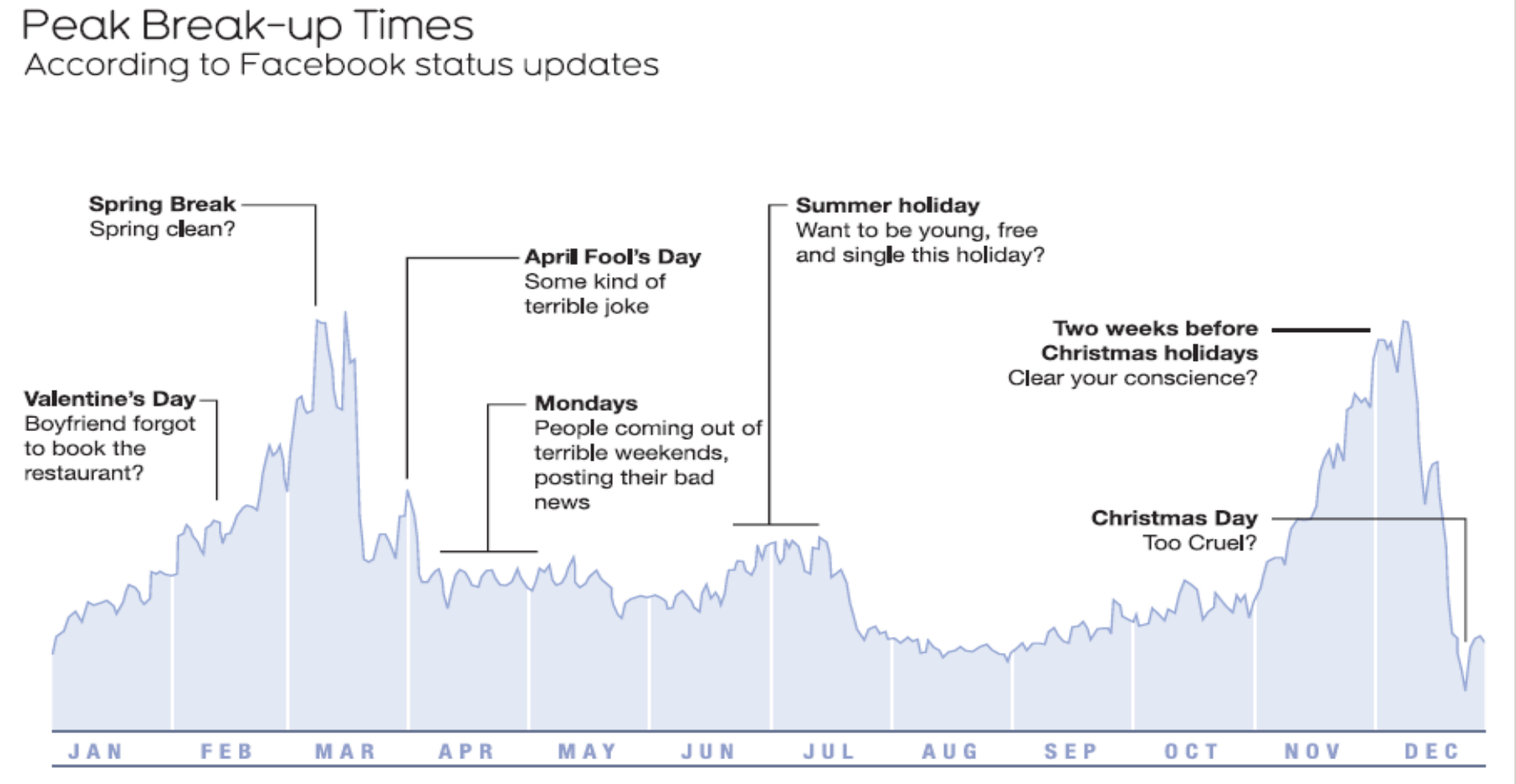
Show, Don’t Tell!



May not be a good advice!

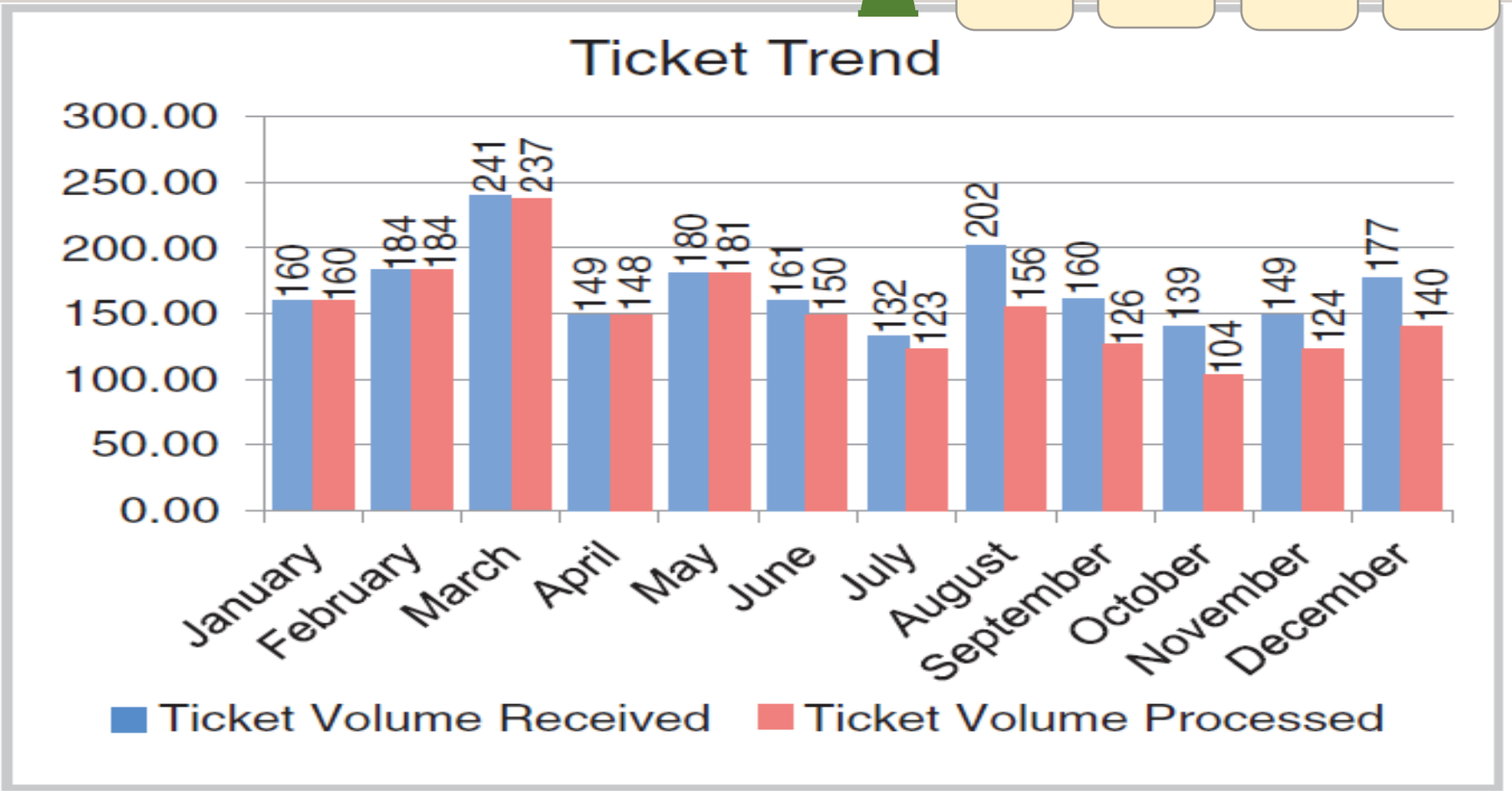
5 4 3 2 1 0

Guess What's that?



Text is your friend.

5 4 3 2 1 0



Show, Don't Tell.

Please approve the hire of 2 FTEs

to backfill those who quit in the past year

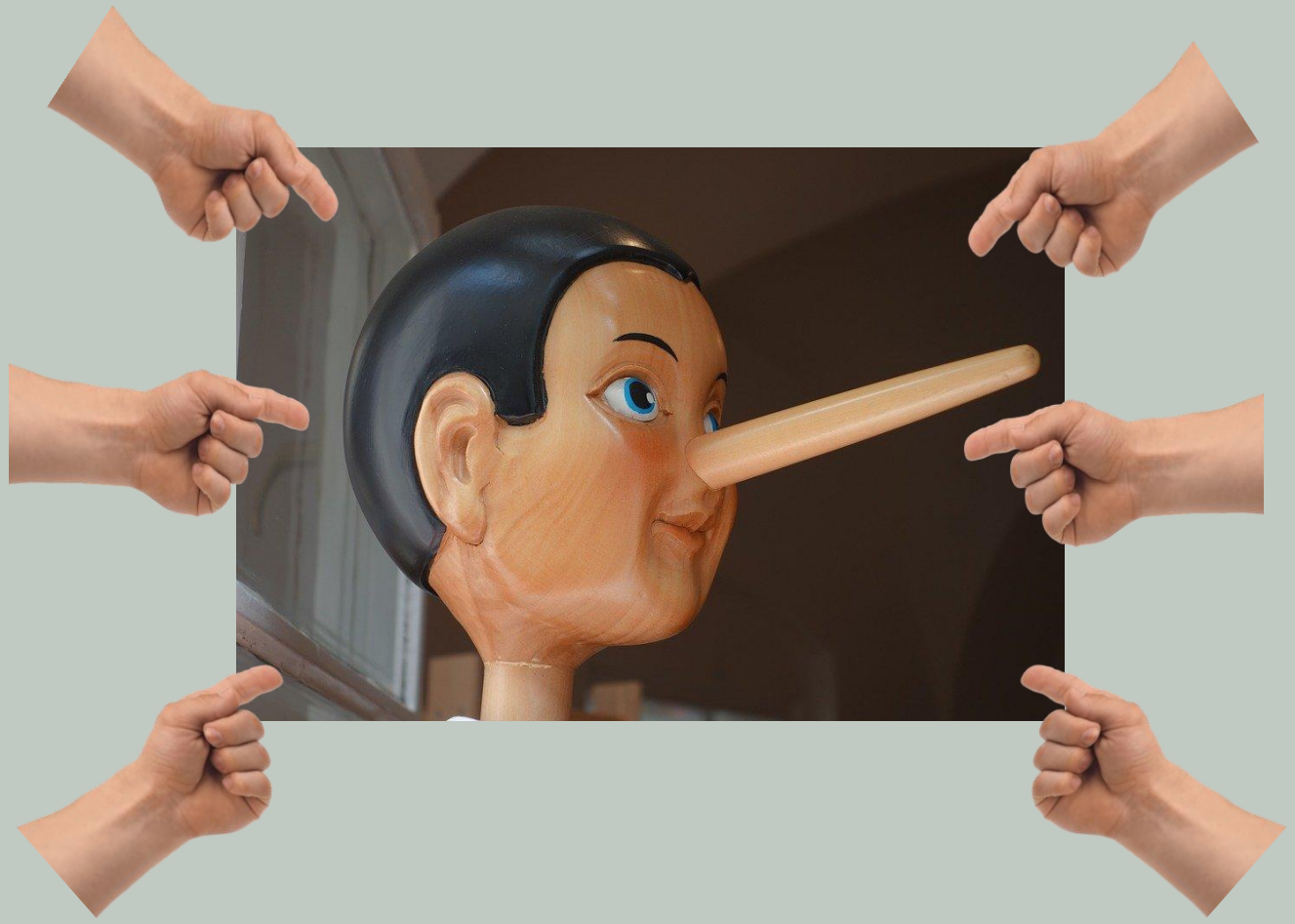
Ticket volume over time

Data source: XYZ Dashboard, as of 12/31/2014 | A detailed analysis on tickets processed per person and time to resolve issues was undertaken to inform this request and can be provided if needed.

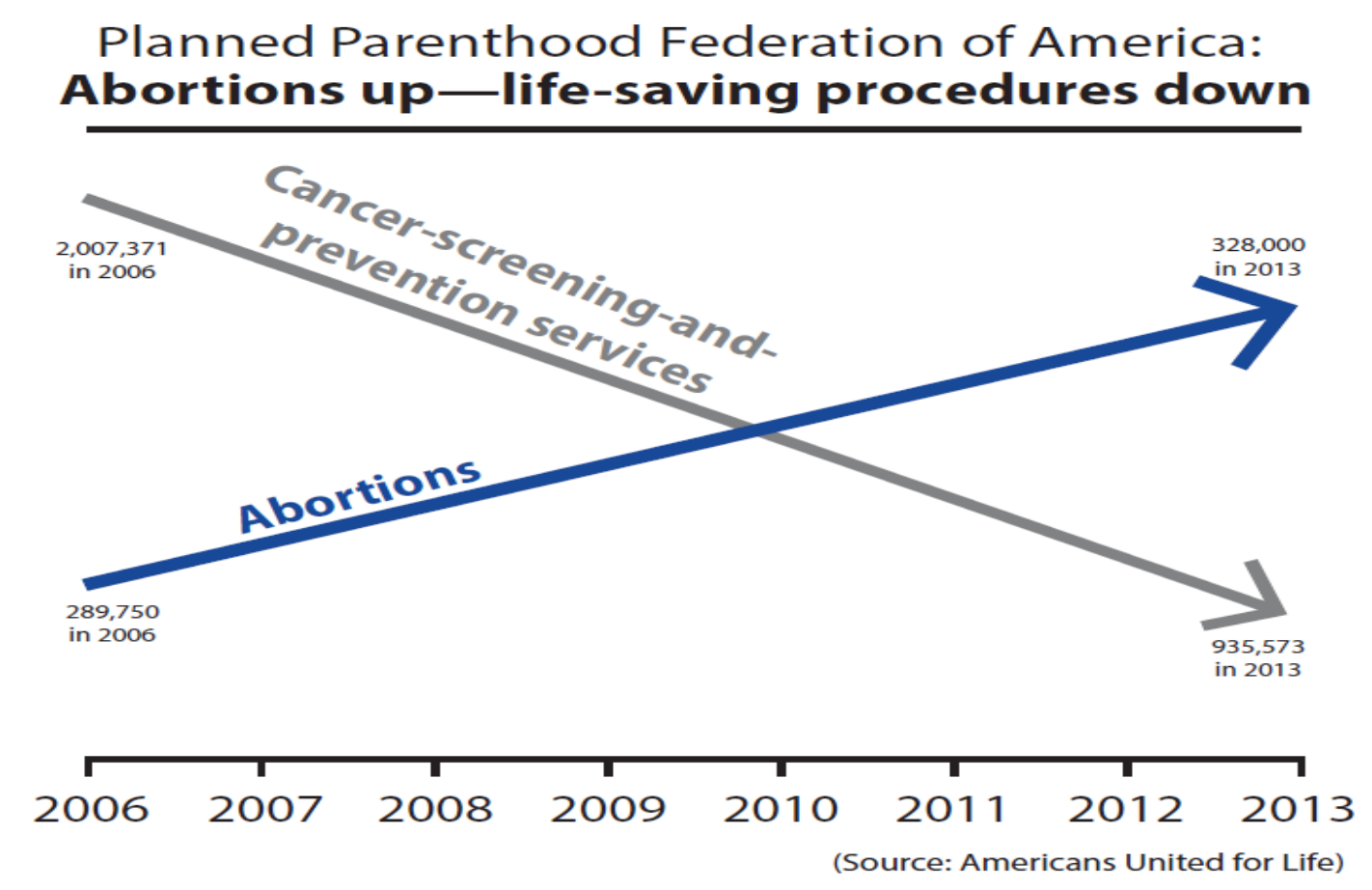
Tell → Show → Why

02

Catch the Lies, if you can!

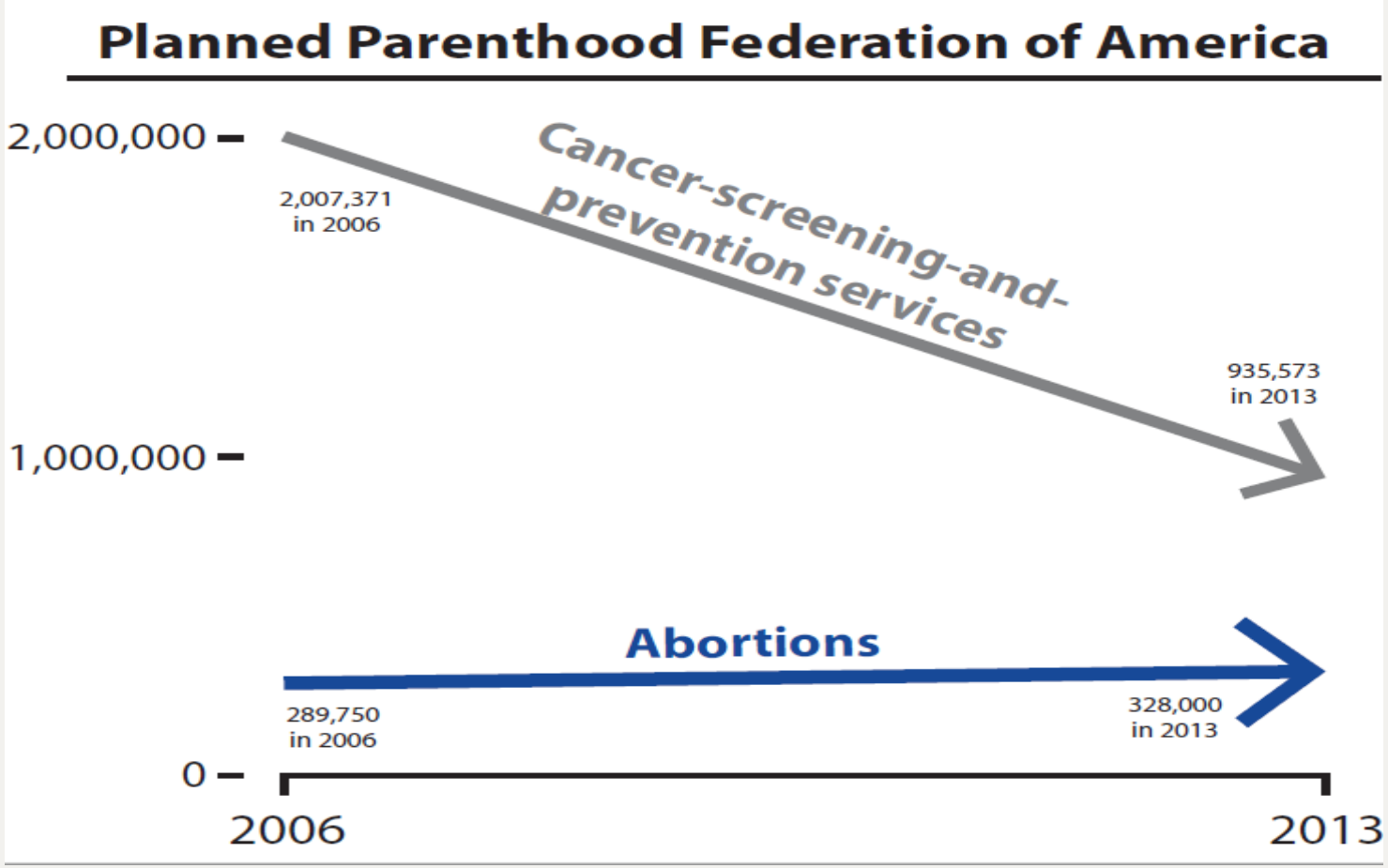


Is the chart lying?

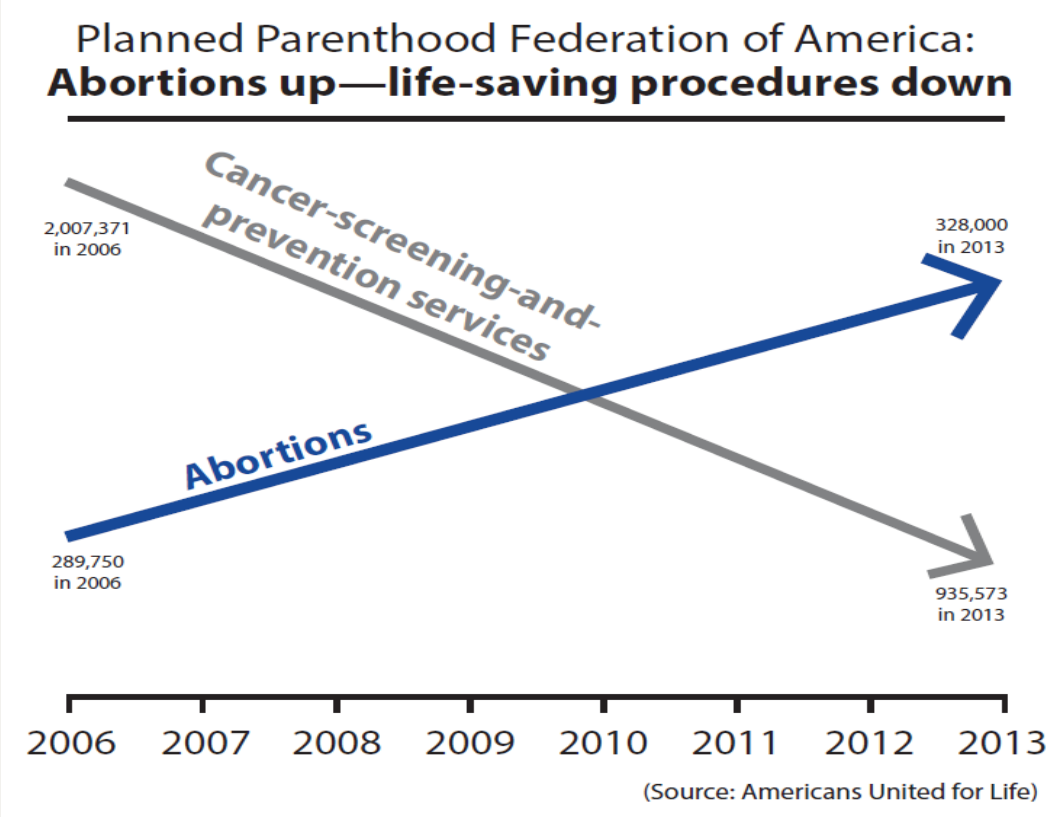


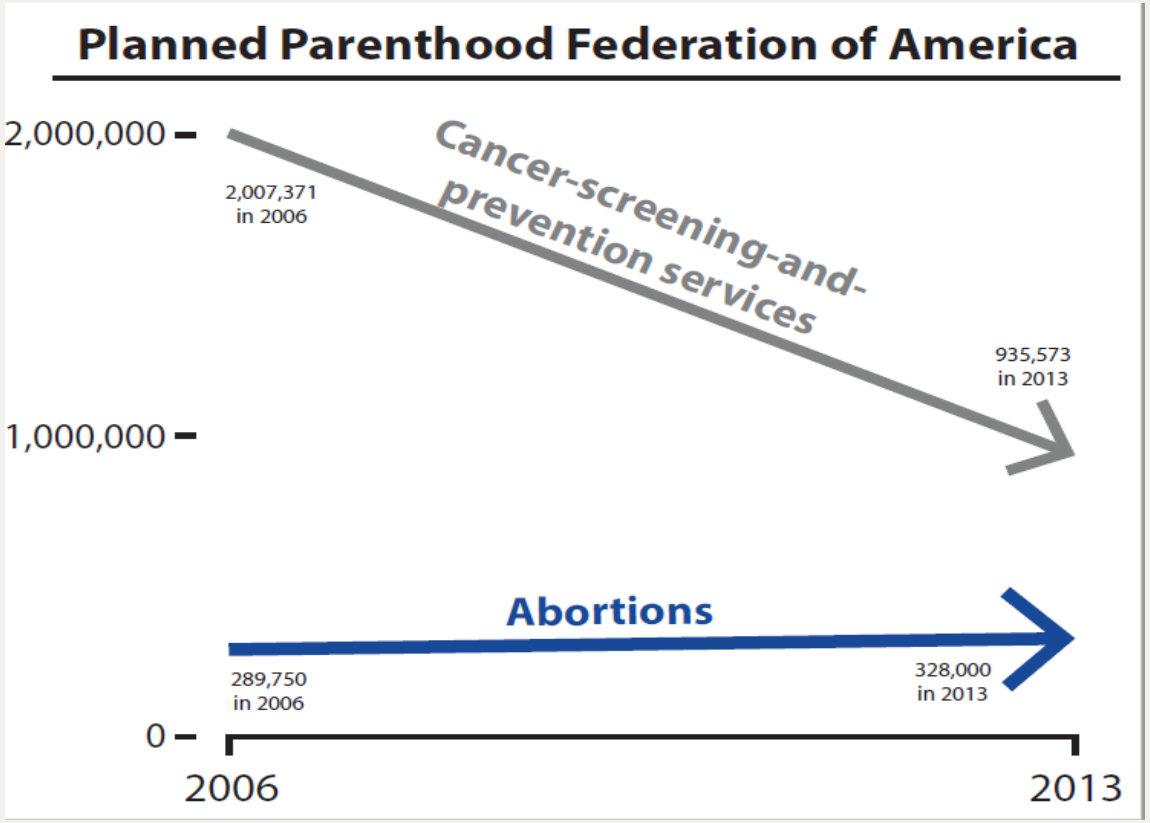
5 4 3 2 1 0

Did you notice?

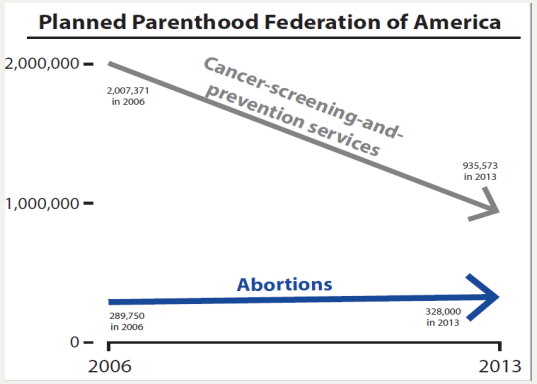


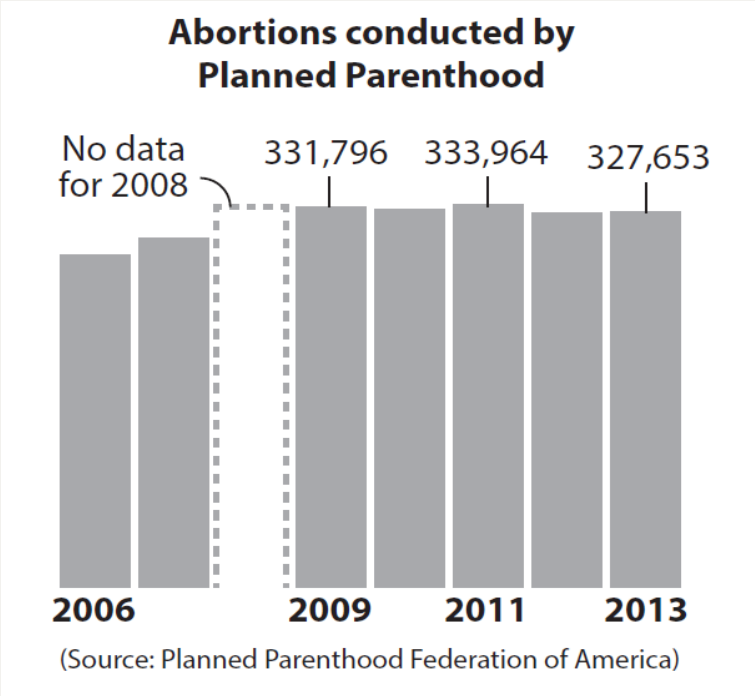
Which one is correct?

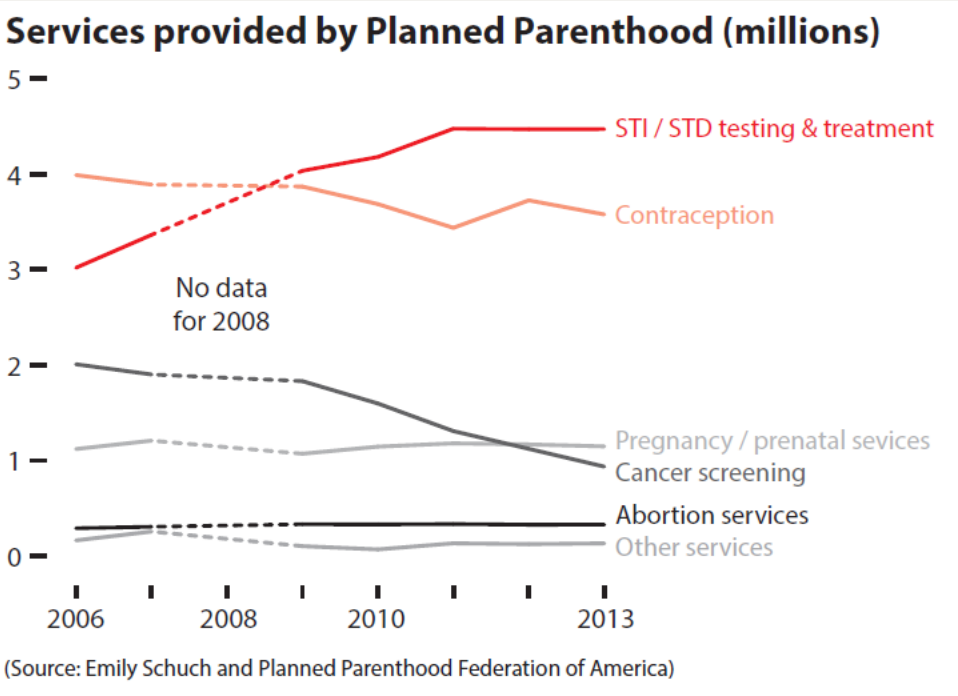




This is not complete yet!



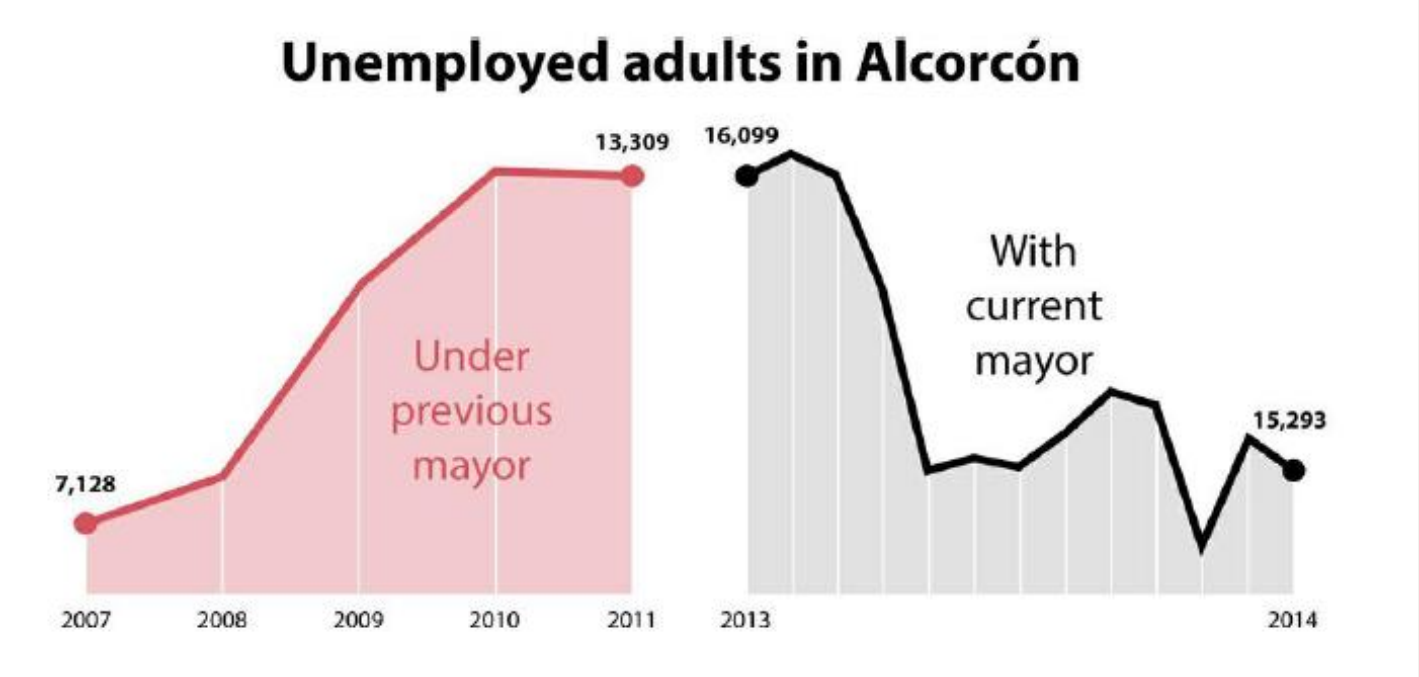




National guidelines for the frequency of cervical cancer screenings were officially changed in 2012, but the American College of Obstetricians and Gynecologists started advising a reduction of frequency in 2009.

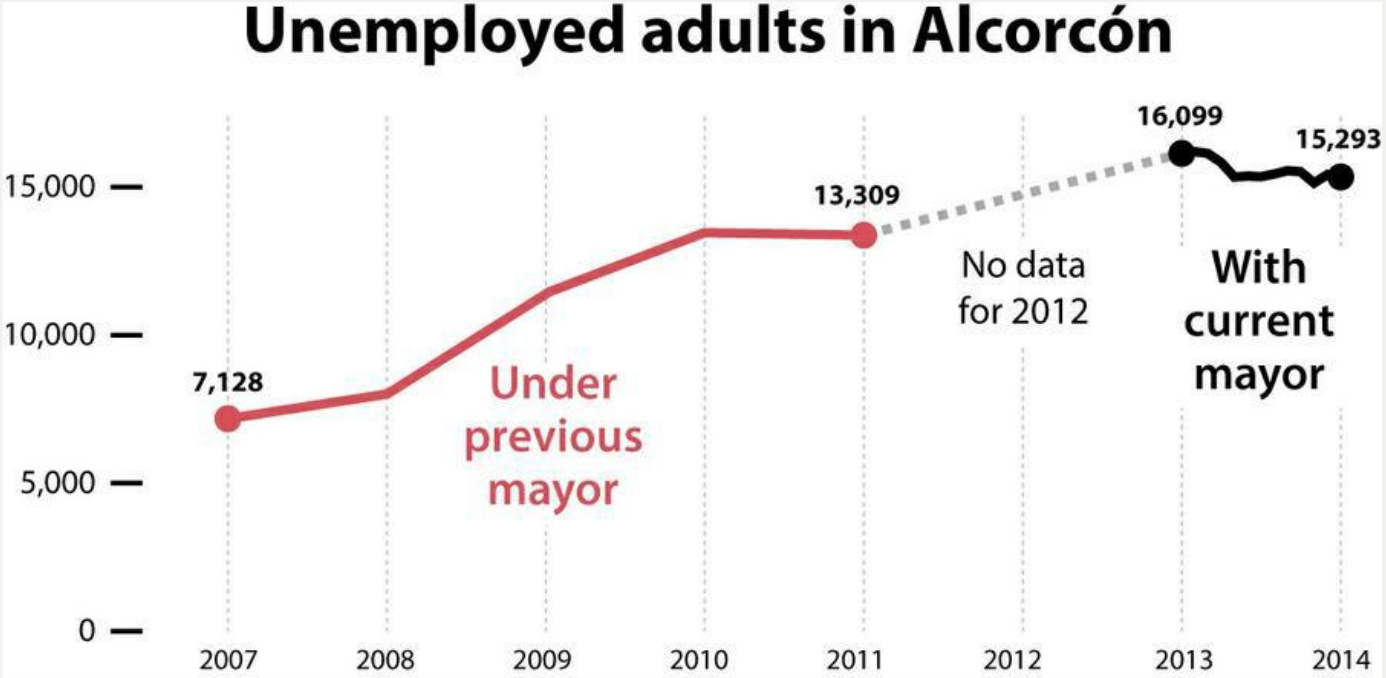
Previously, women received a pap smear to screen for cervical cancer once a year, but it is now recommended that pap smears should be performed every three years.

Is the chart lying?

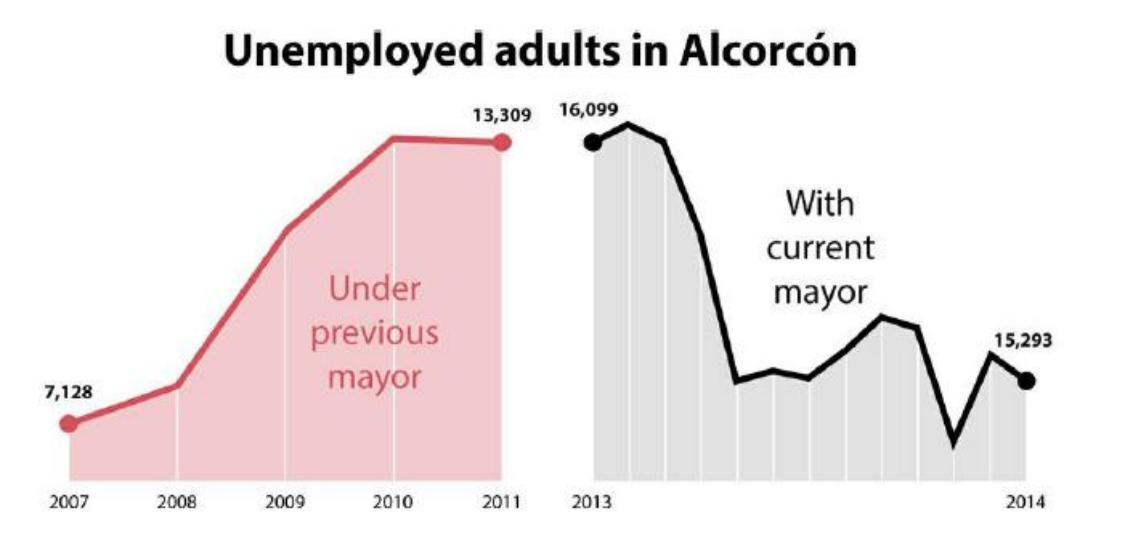


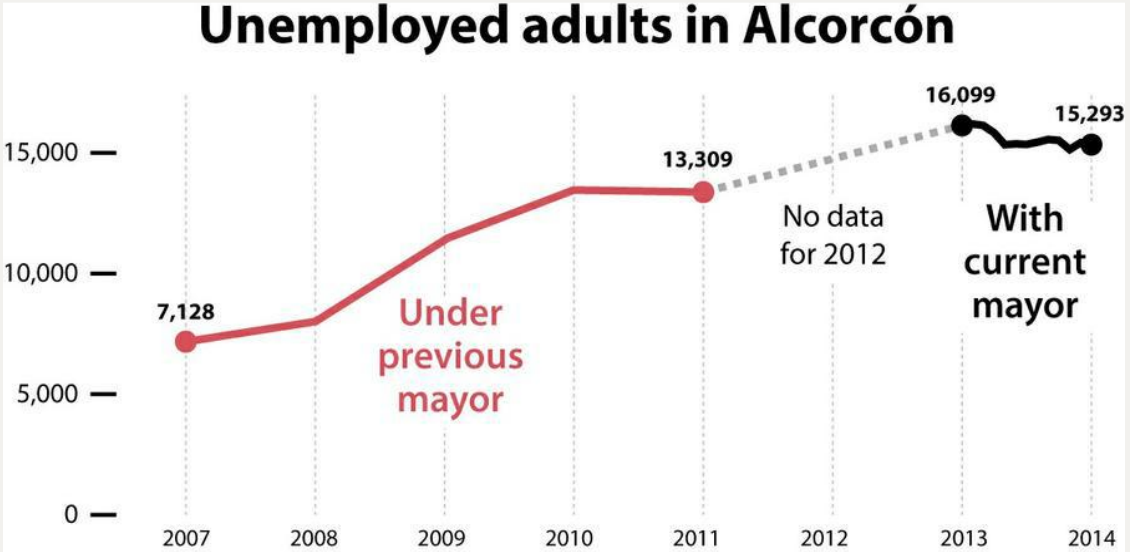
5 4 3 2 1 0

Did you notice?



Which one is correct?

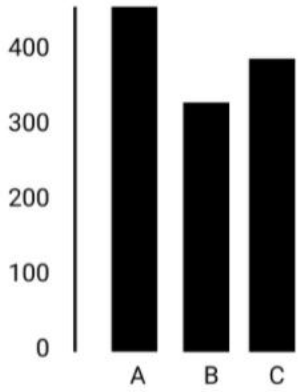




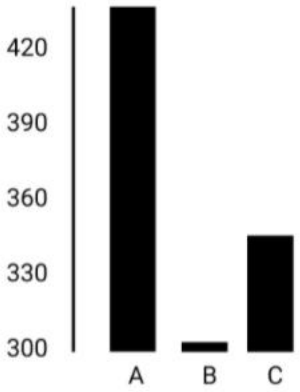
03

Know The Rules!

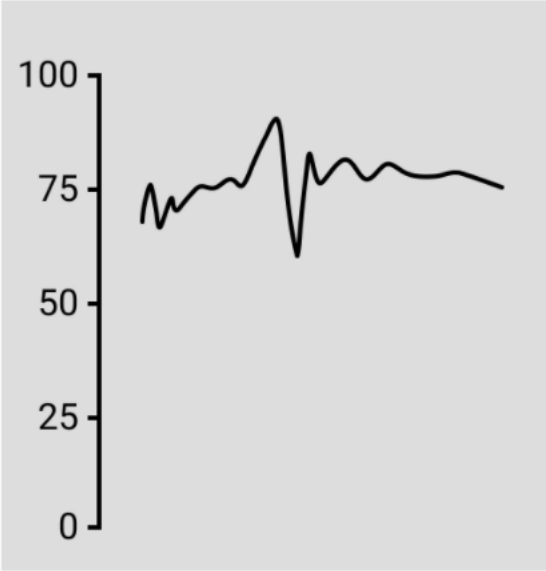
Zero/Non-Zero Baseline

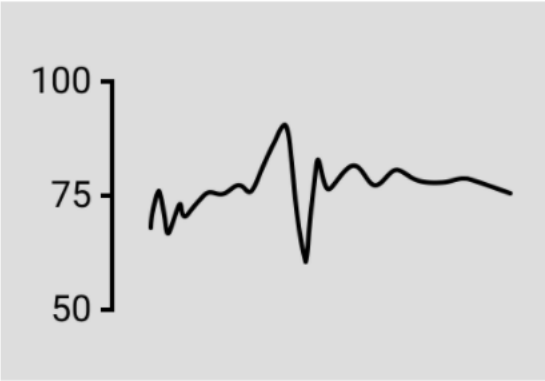


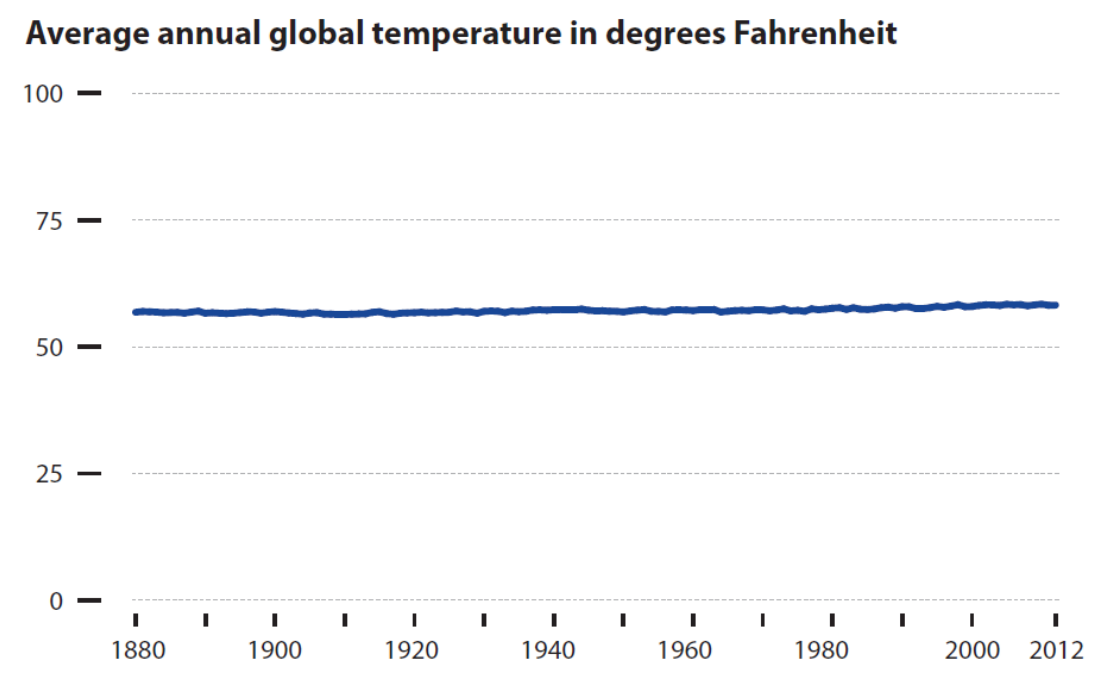
Good. Y-axis starts at 0 and has natural increments



Bad. Y-axis starts at 300, exaggerating the difference among columns. Intervals are awkward

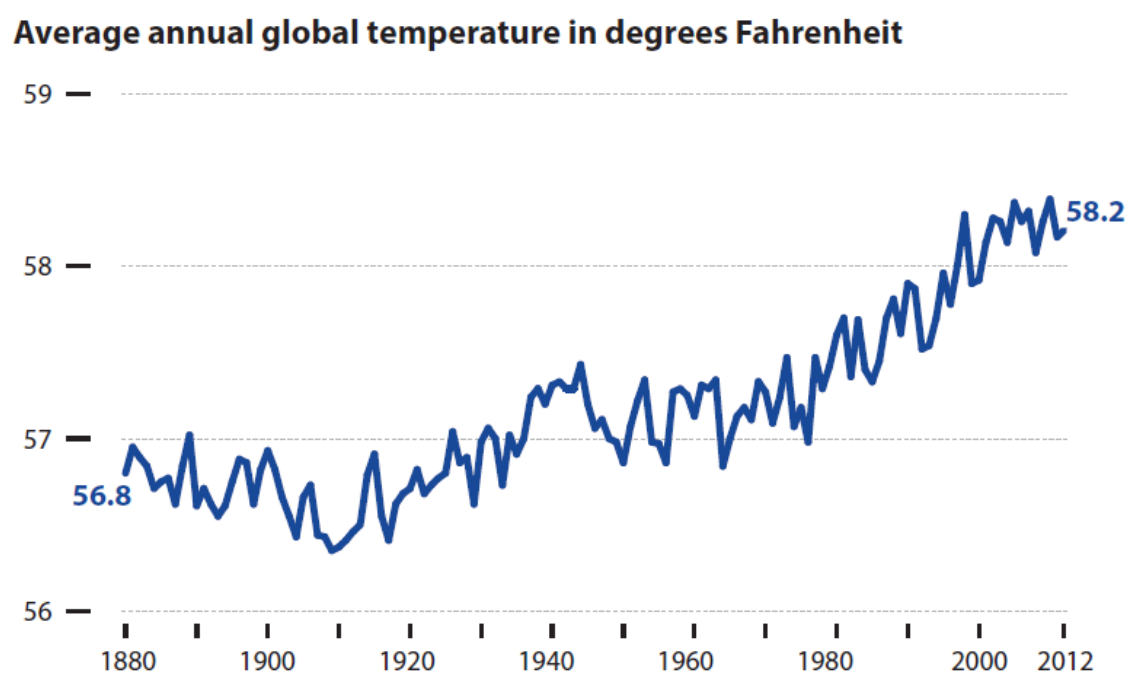




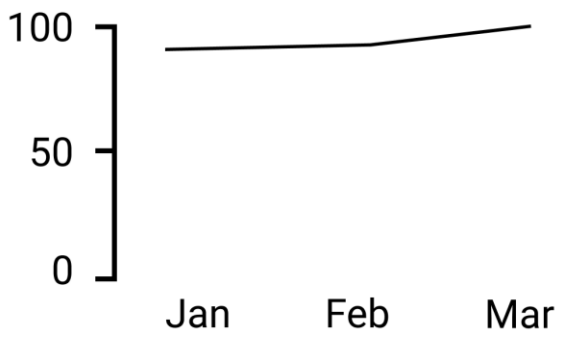


The only climate change chart you need to see

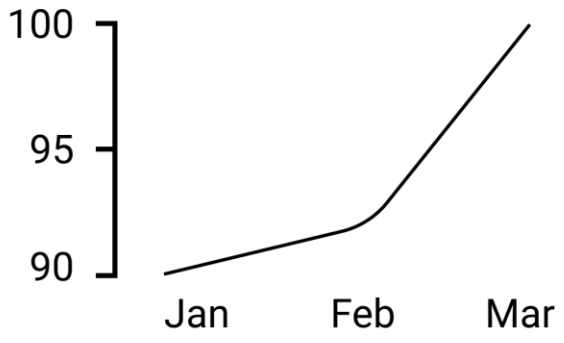
– *National Review* magazine, 2015



More on climate change:
https://www.bbc.com/news/science-environment-46384067



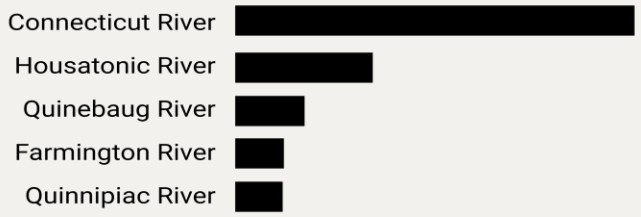
Acceptable, but starting the vertical axis at zero obscures changes in values



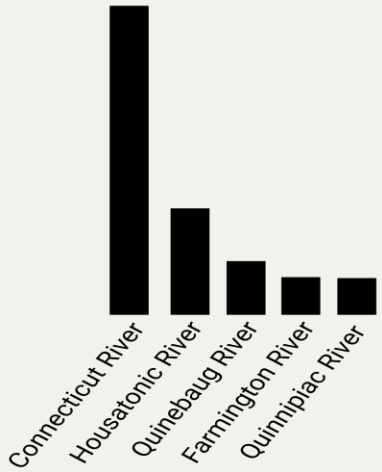
Better: By reducing the vertical axis to match the values, we see change more clearly

Zero/Non-Zero Baseline depends upon nature of information, available space, and the data story

Readability

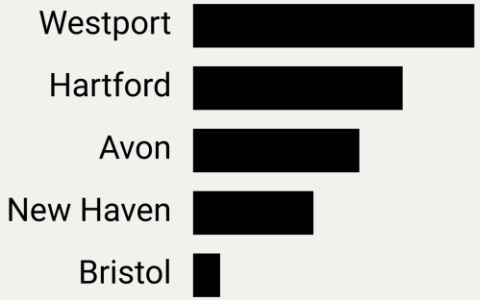


Good



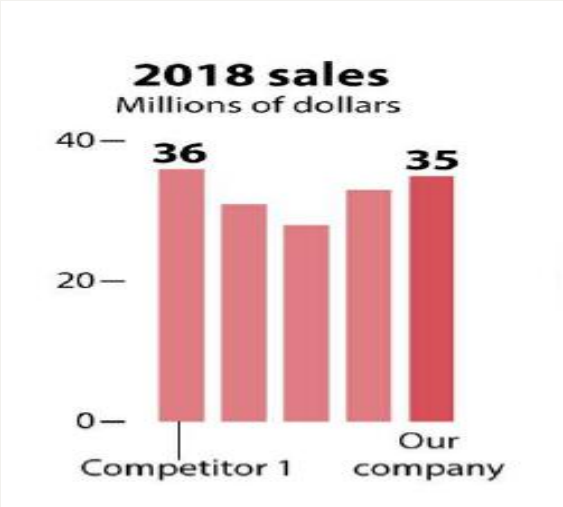
Could be better

Good (Alphabetical)



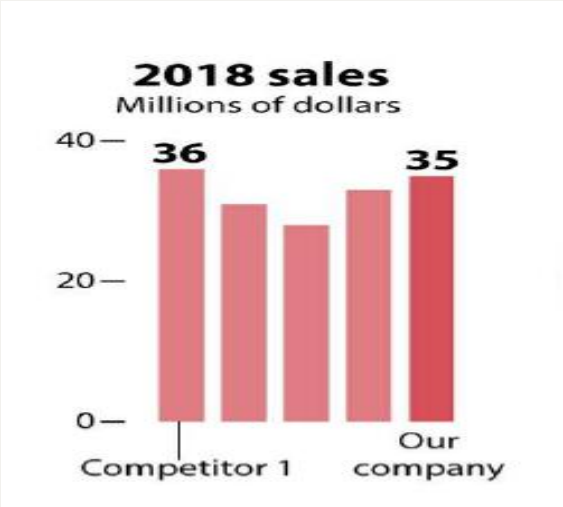
Good (Values sorted)

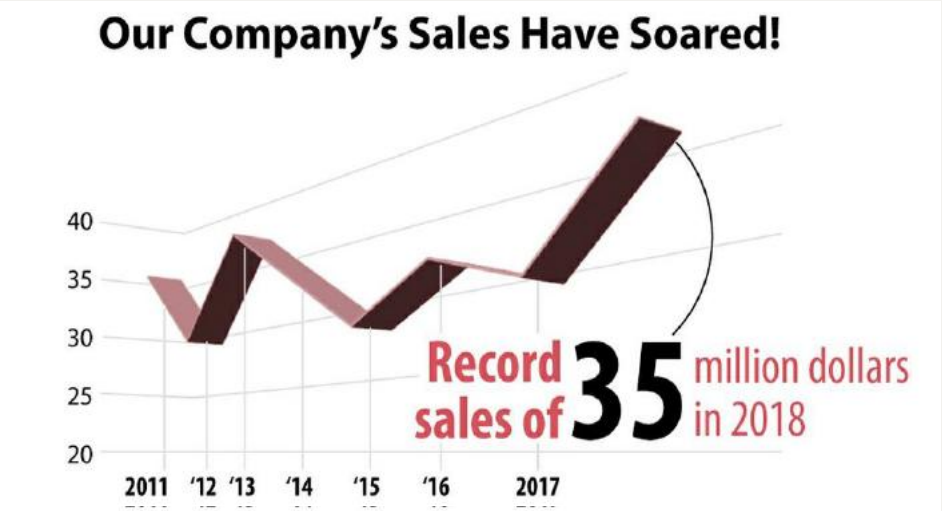
Bad (Random order)

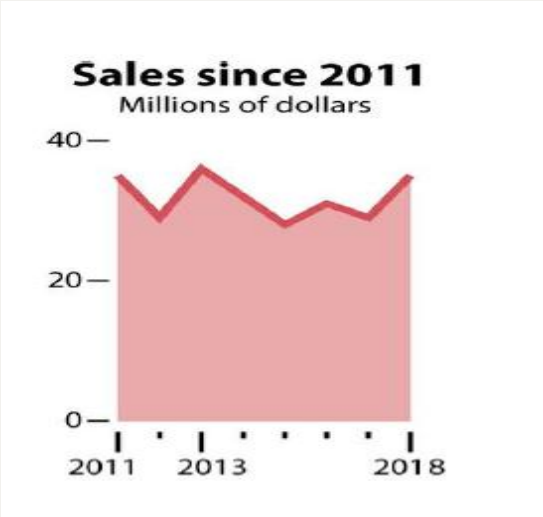


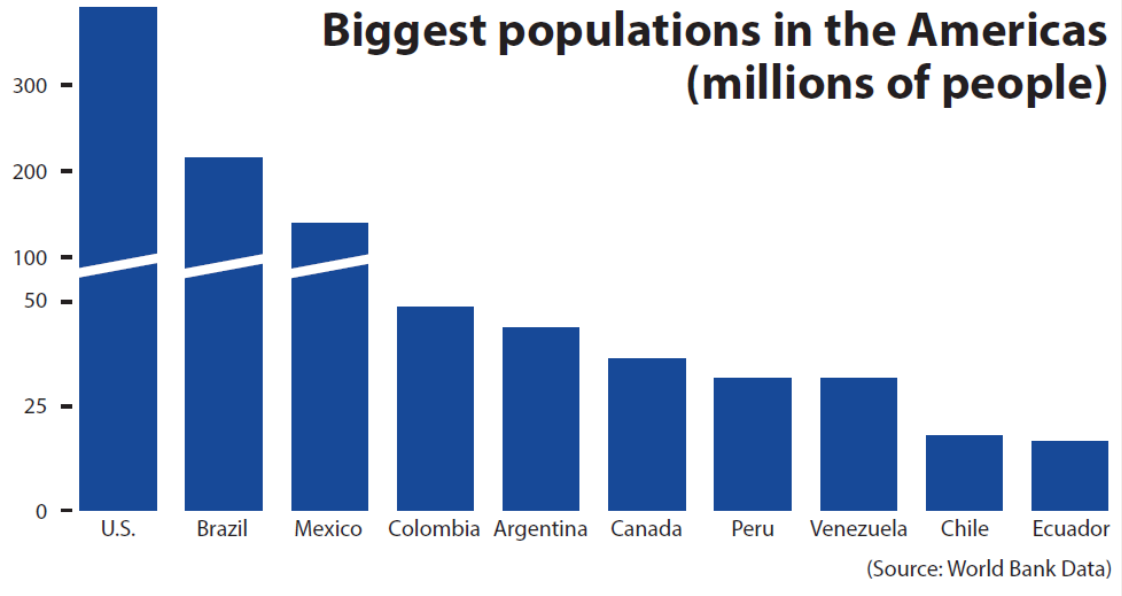
Avoid 3D Plots

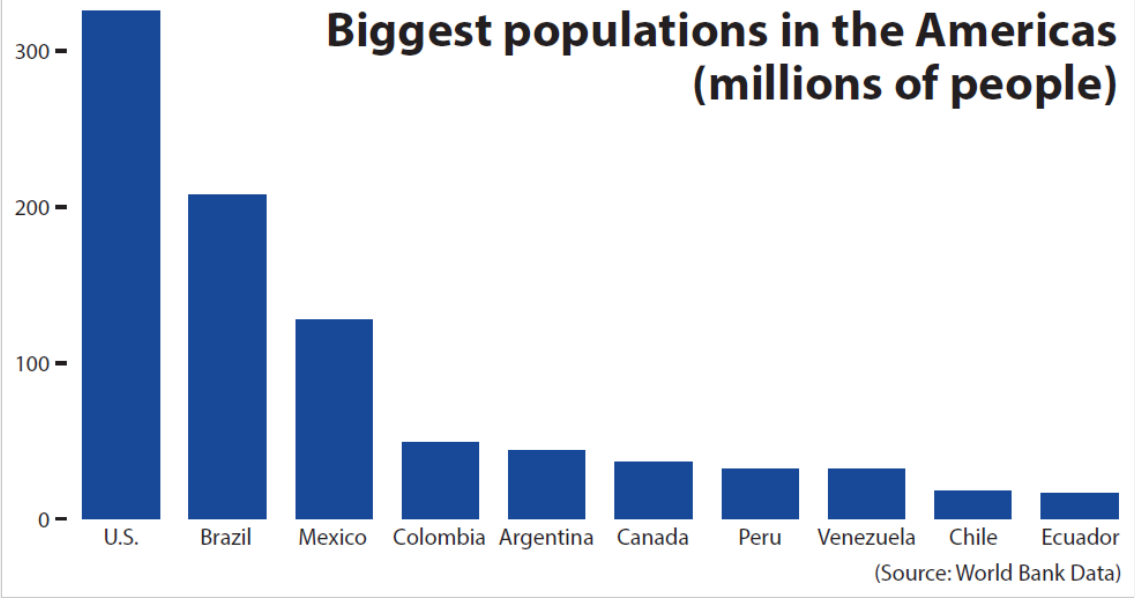
Avoid 3D Plots







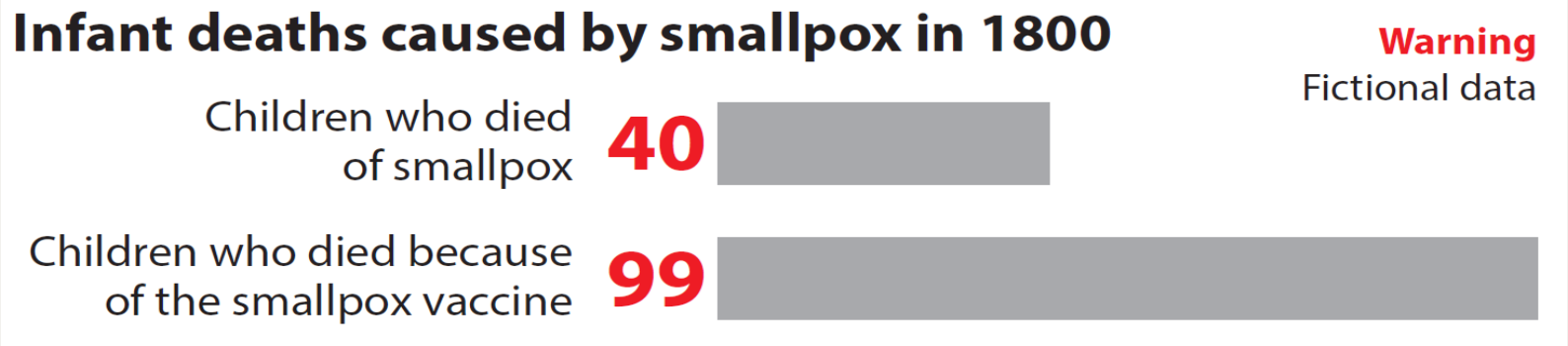


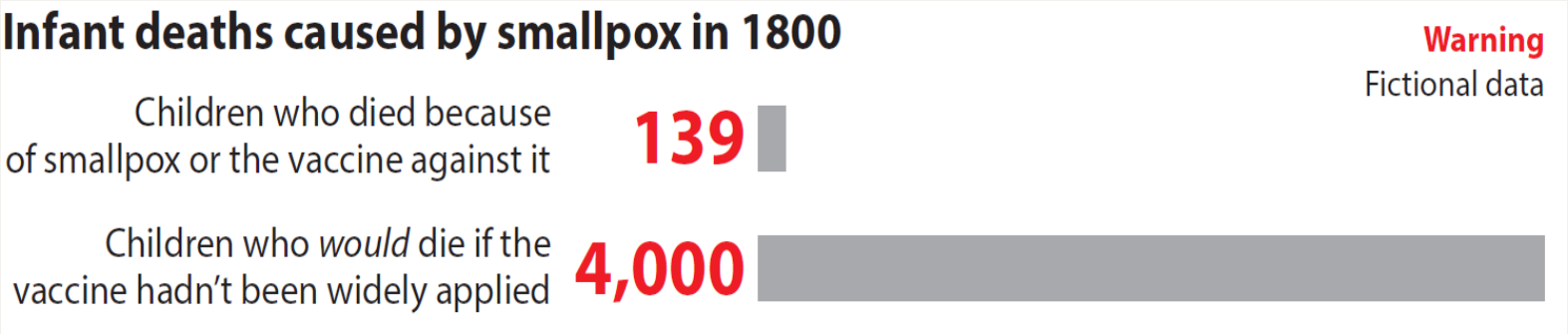


Avoid Truncating

Don't mislead by showing

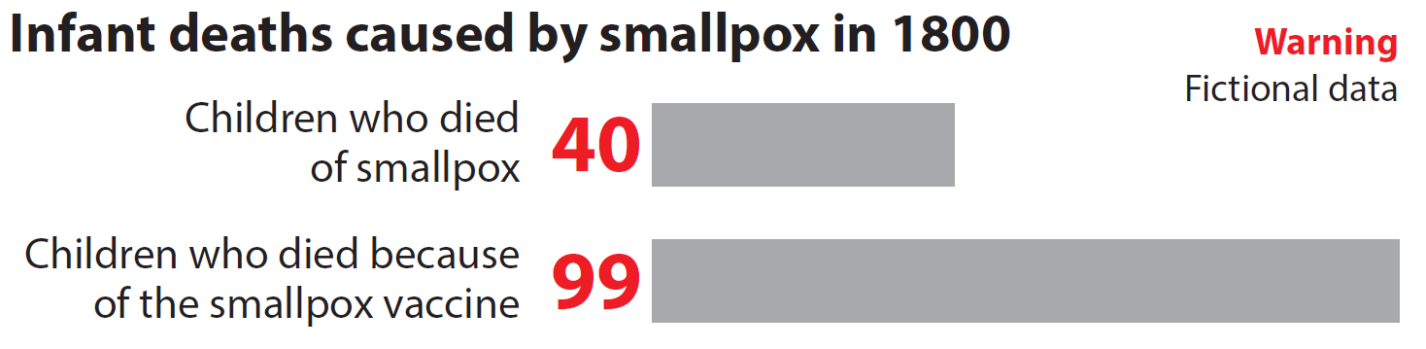
Too Little Information

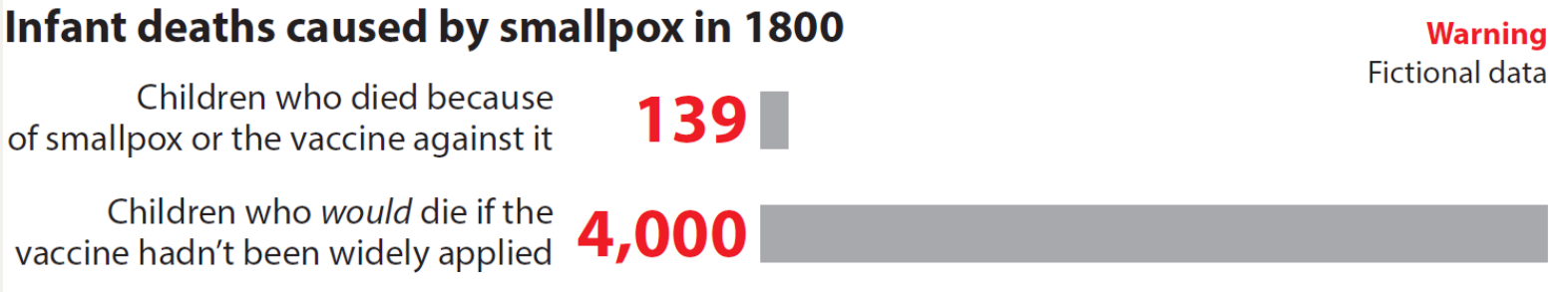


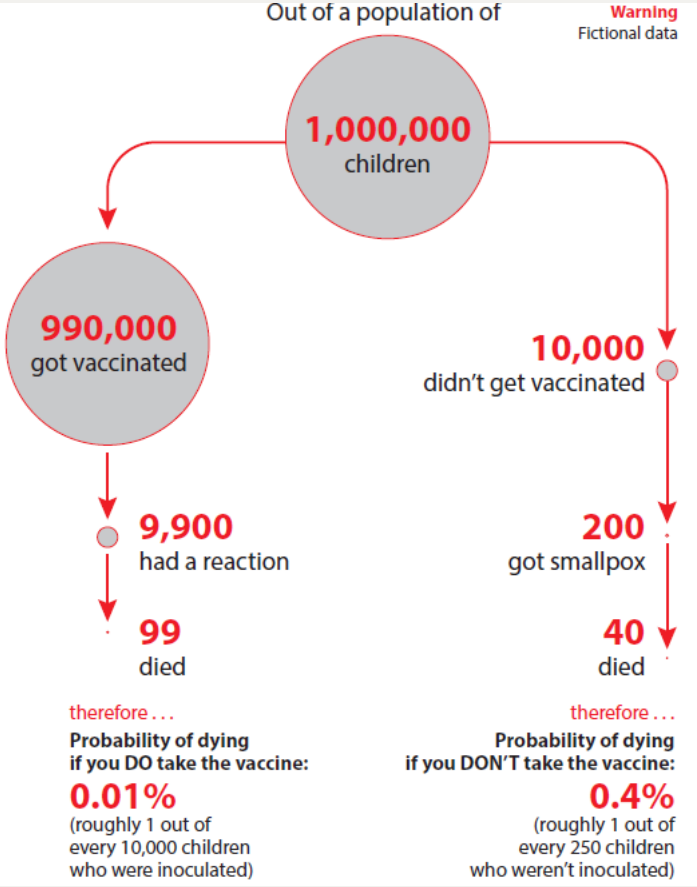


Don't mislead by showing

Too Little Information

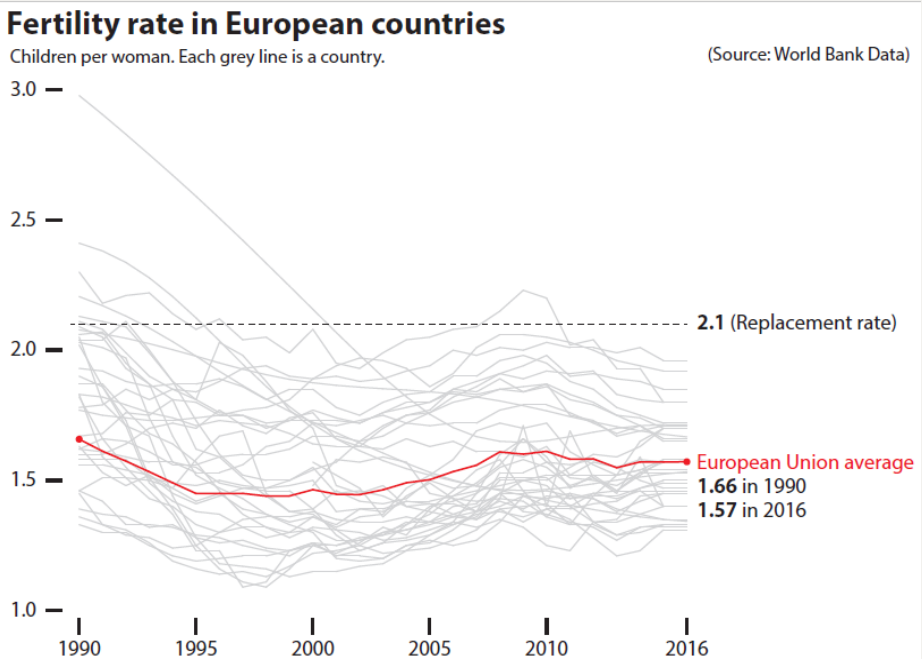


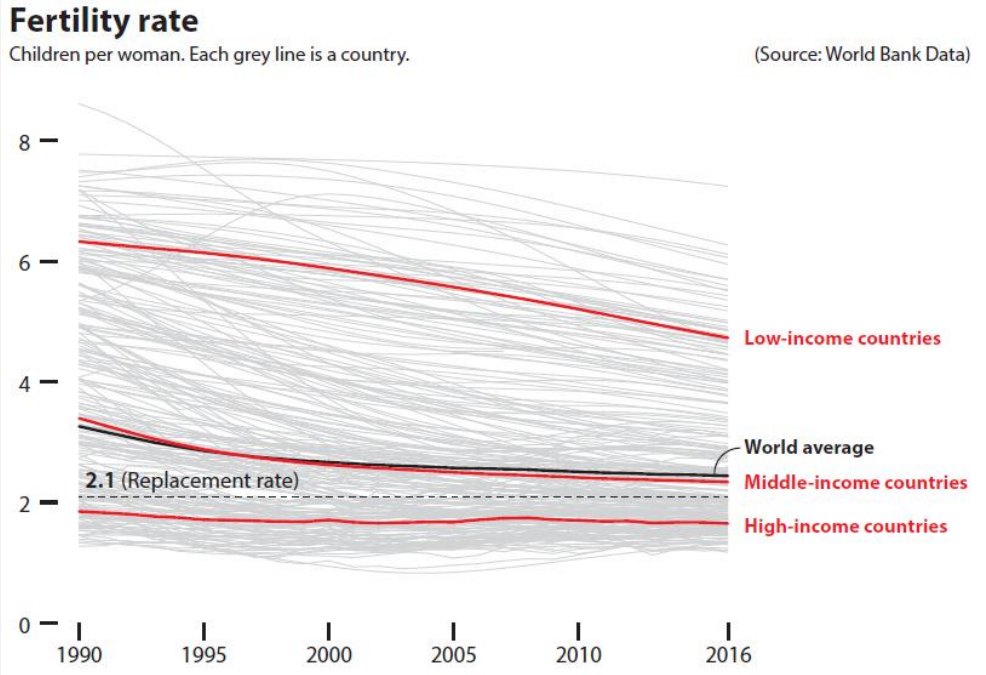


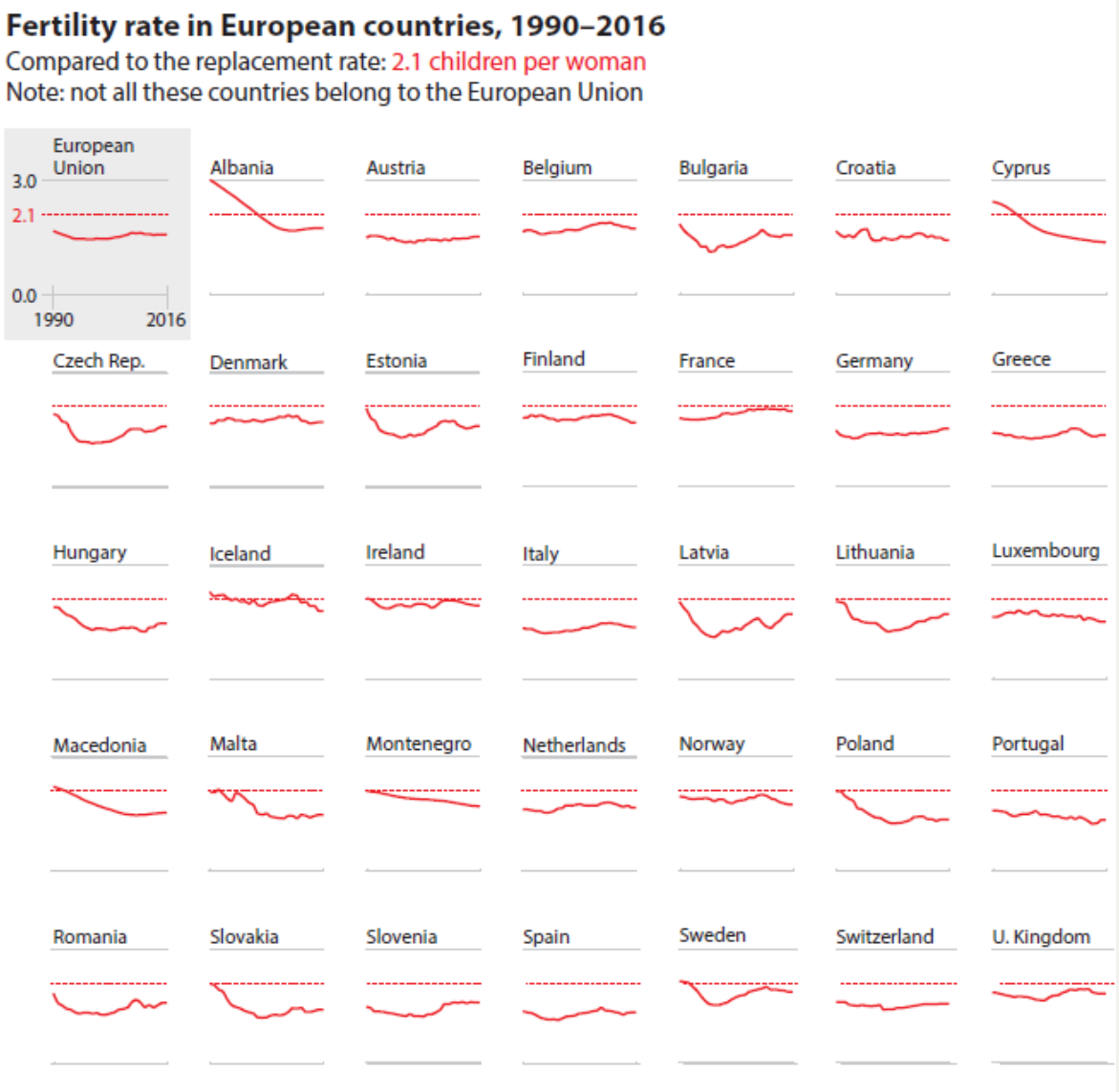


Don't mislead by showing

Too Much Information





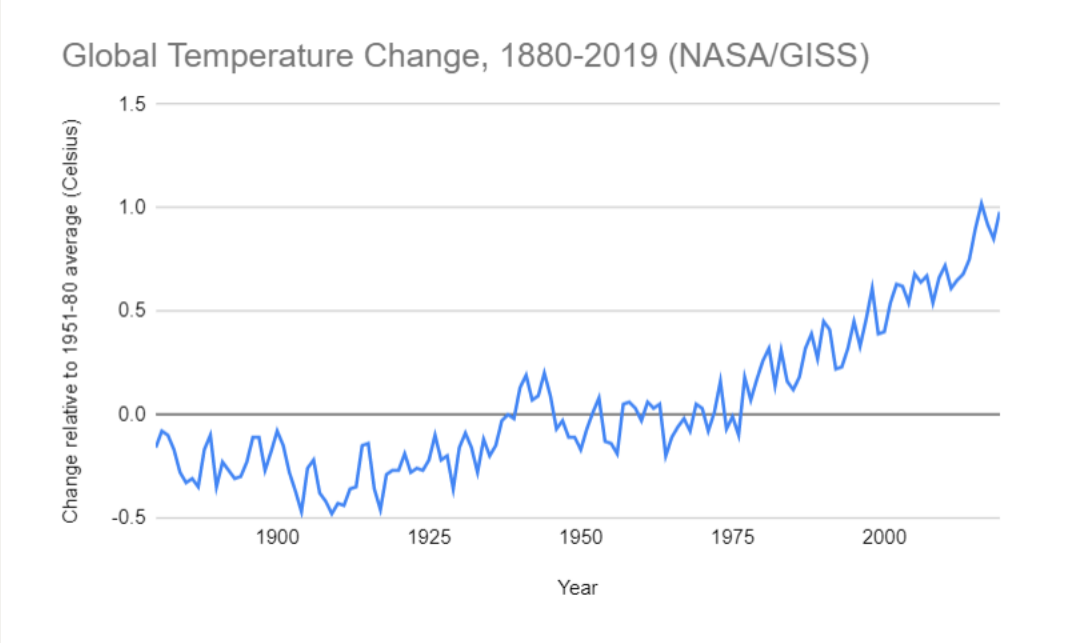


Don't mislead by showing

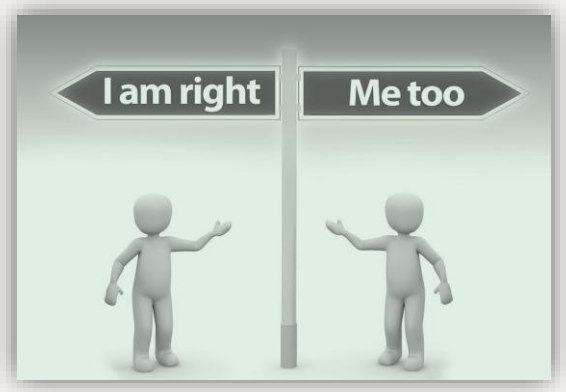
Too Much Information

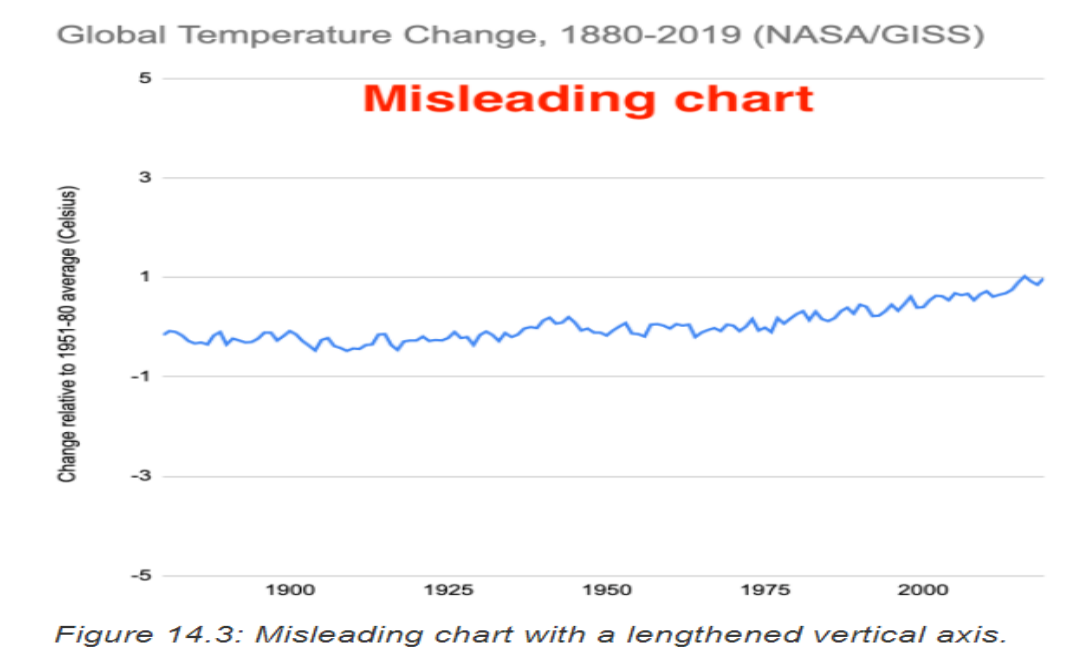
04

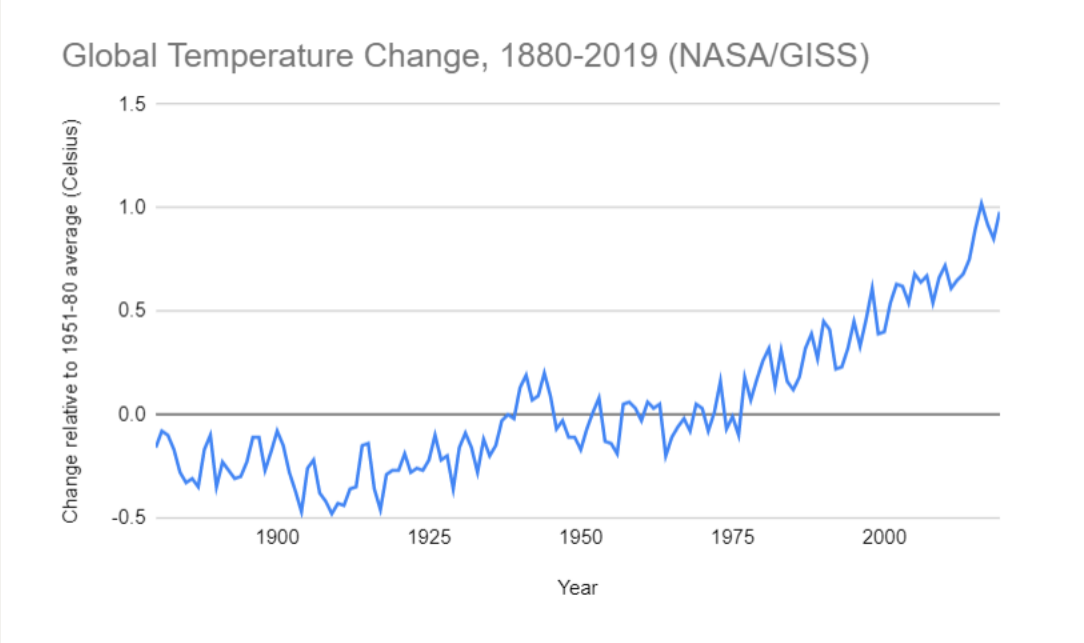
Technically Correct _ yet misleading charts!



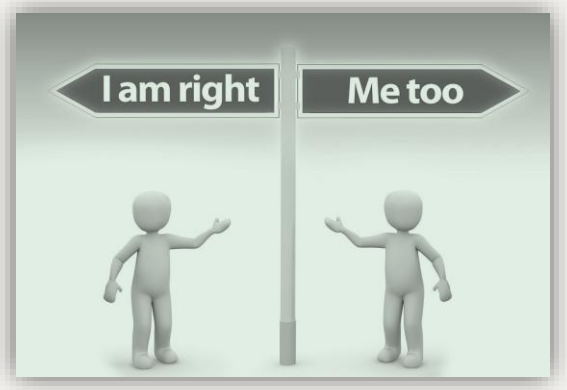
Is it worrisome?

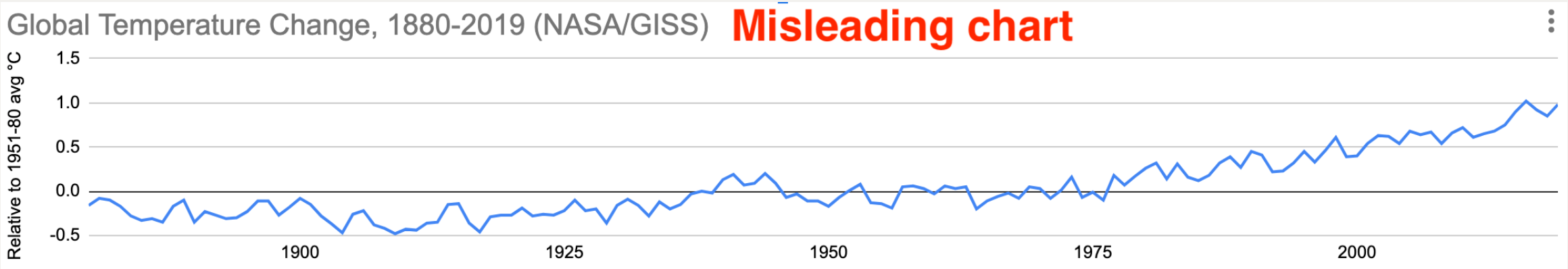




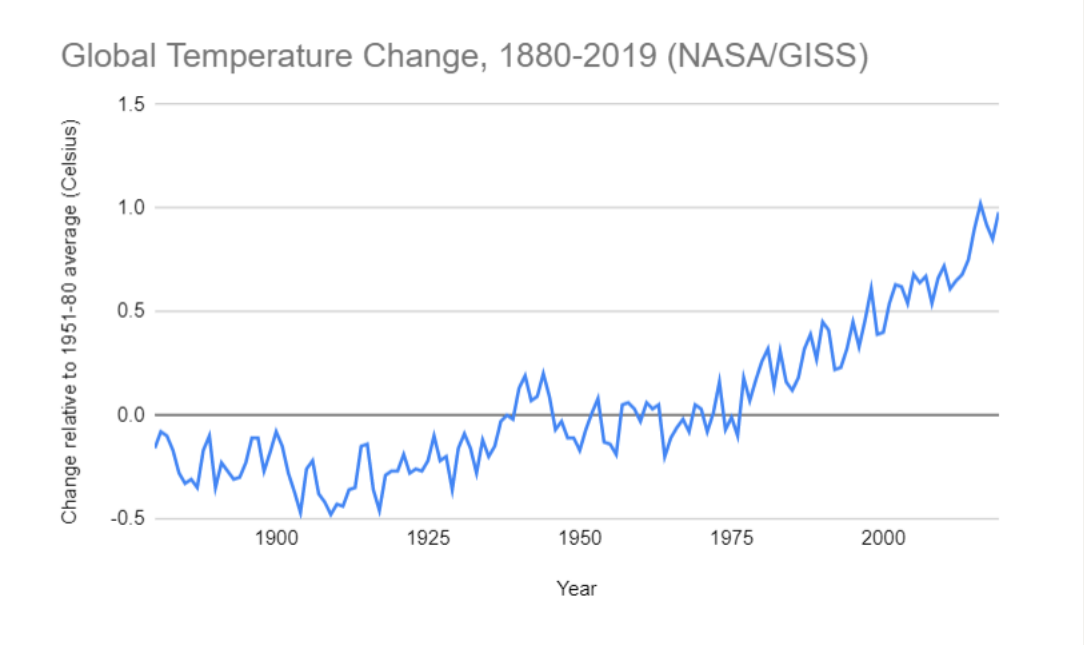


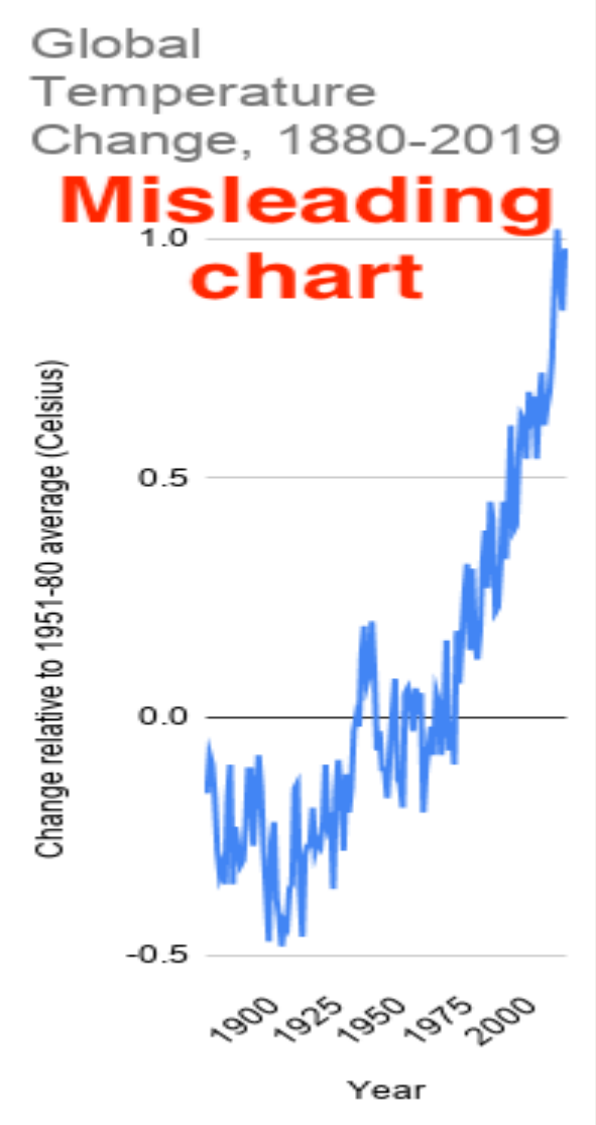
Is it worrisome?

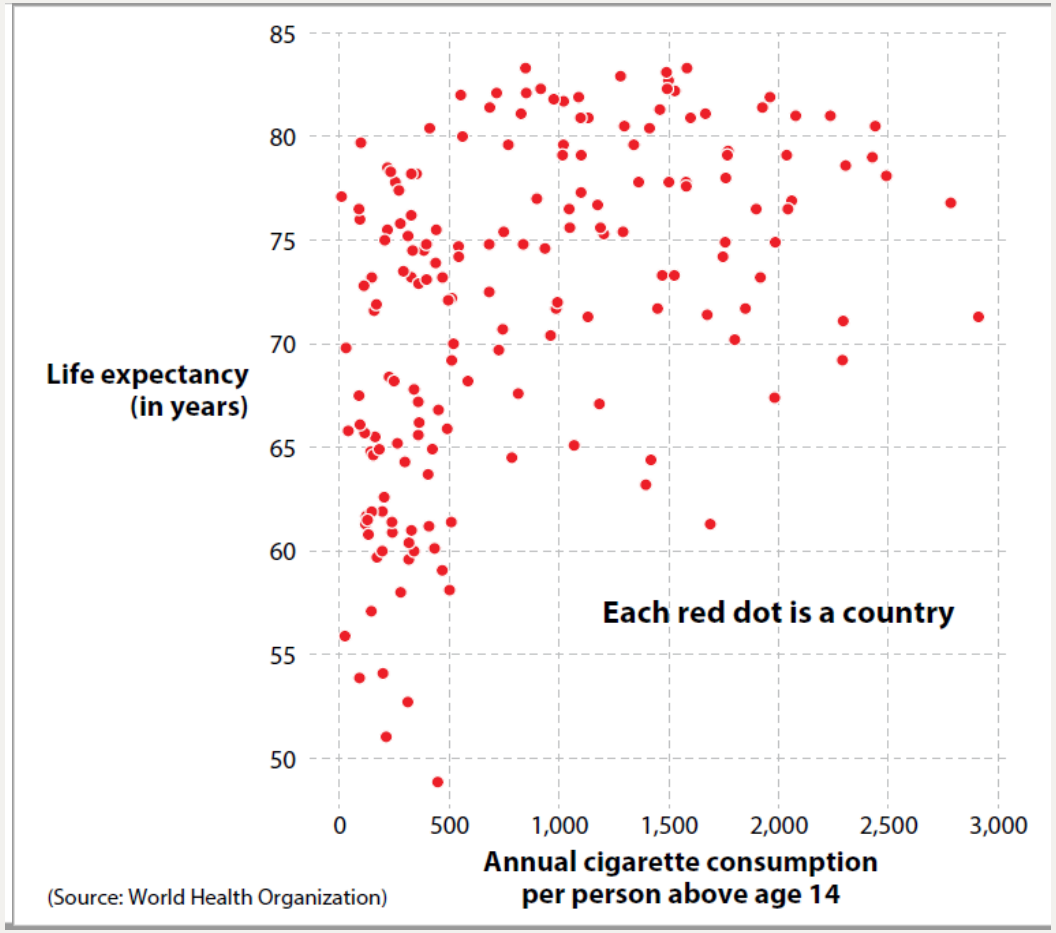




Is it worrisome?



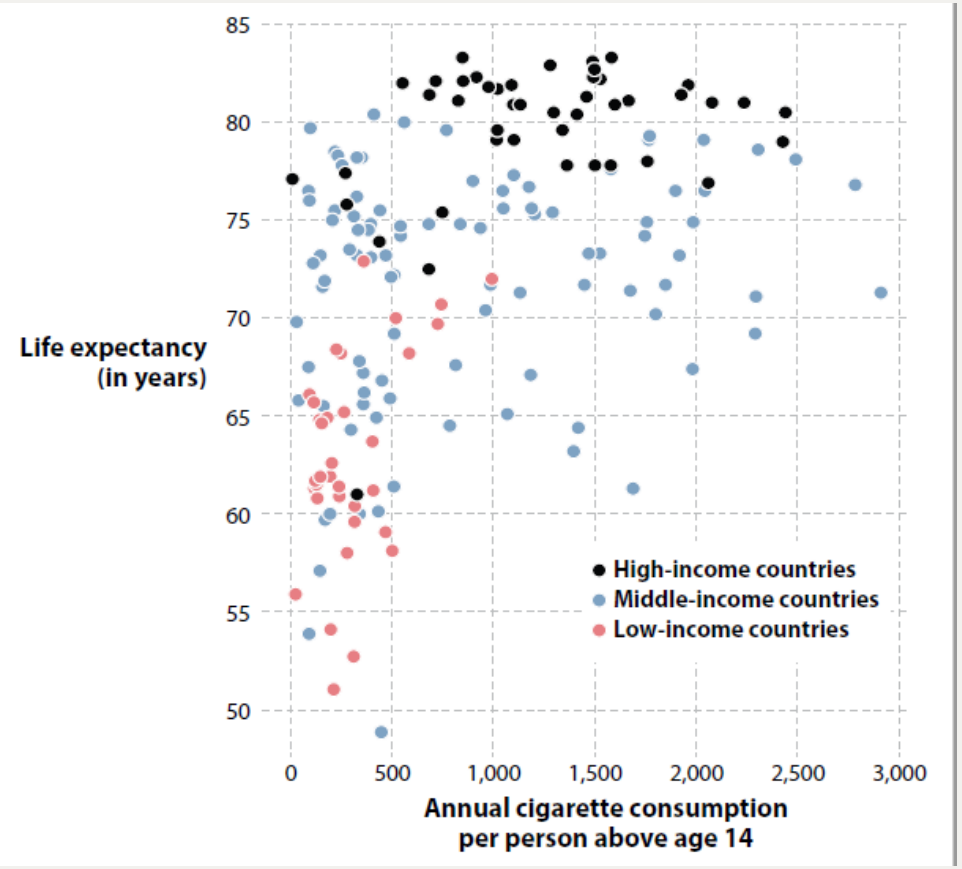




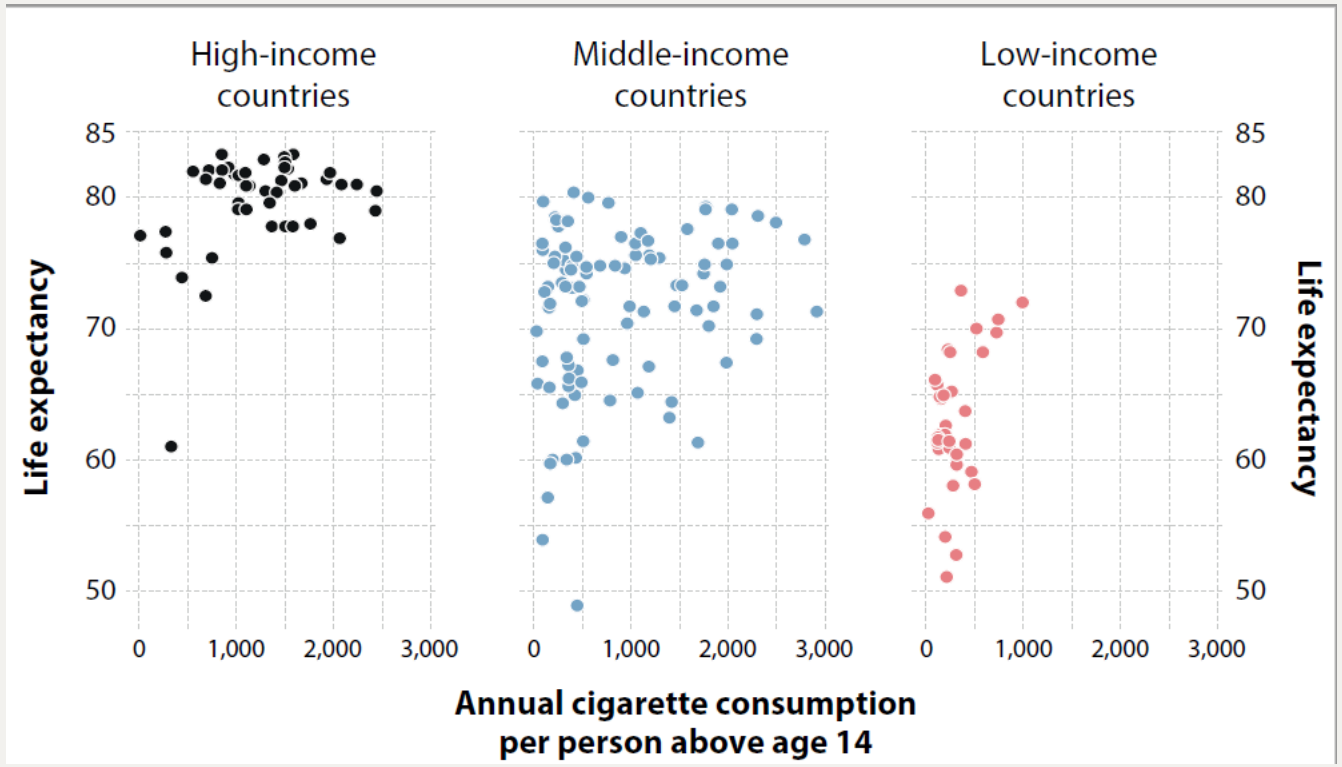
Strange, Isn't it?

5 4 3 2 1 0

"The more cigarettes we consume, the more we live"

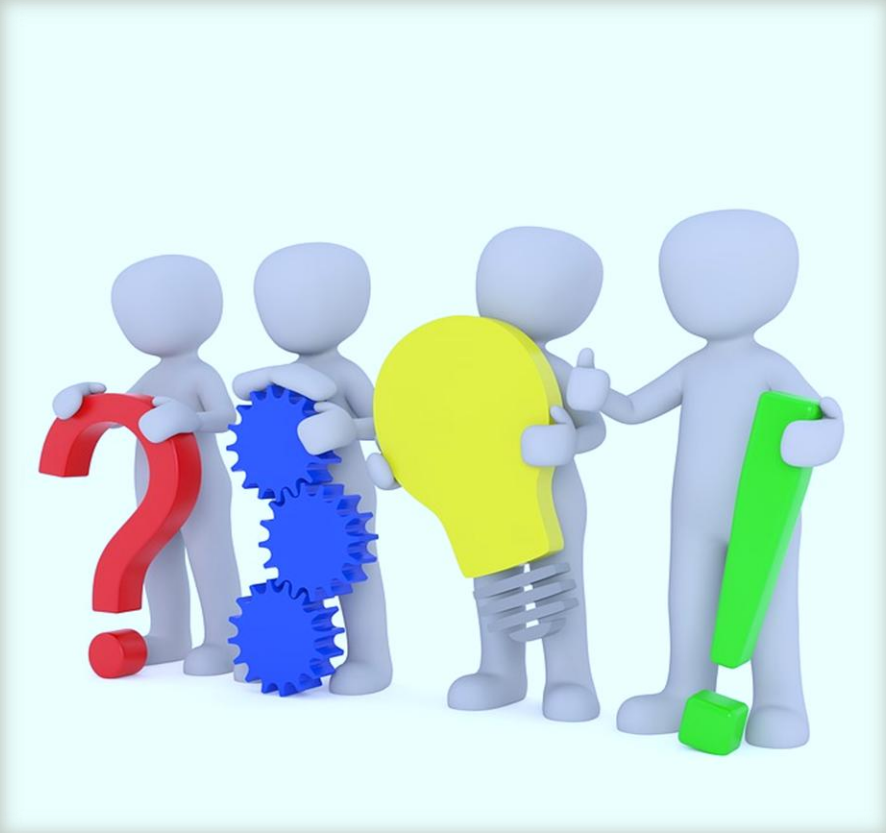


Did you see it now?



05

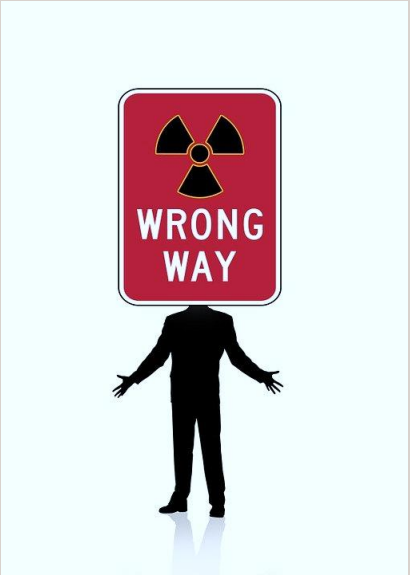
Closing



Don’t let anyone deceive you by a chart

A chart can be misleading

- ✓ If it is poorly designed
- ✓ If it used wrong data
- ✓ If it shows too little or too much
- ✓ If it conceals or confuses uncertainty
- ✓ If it suggests misleading patterns



I tried to make you a better at …

Reading Charts We had learnt it passively for some charts	**Catching Lies in chart** You've learnt to save yourself from mis-information to some extent
Chart Designing Rules Some Don's & Don'ts, we have discussed	**Seeing the complete picture** Asking for more/finer information, if needed

Talk Glimpse

Storytelling with Data

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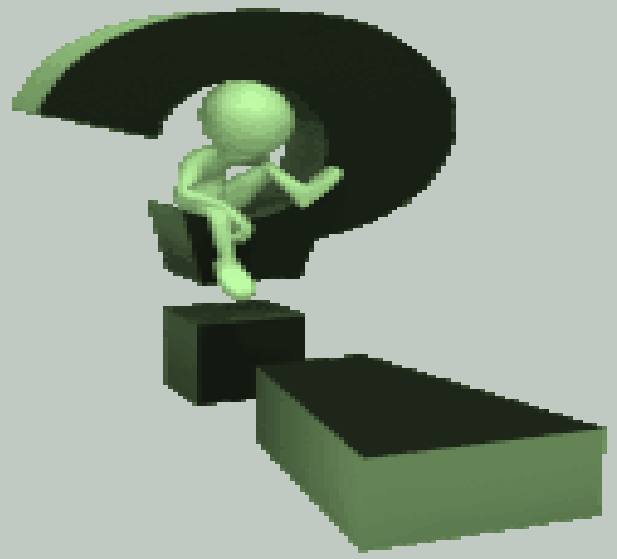
Questions!



Thank You

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